

Fluorescence Confocal Microscope & TCSPC System Manual

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Contents

1	Introduction	4
2	System Overview	5
2.1	Detectors	7
2.2	Photon-counting	10
2.3	High-speed Analog Scanning	10
3	Confocal Microscopy	11
3.1	Infinity-Corrected Optics	11
3.2	Simulation of Confocal Principle	12
4	Time-Related Single-Photon Counting (TCSPC)	13
4.1	Working Principle	13
4.2	Reverse Start-Stop Configuration	15
4.3	Delay Line Configurations and Optimal TCSPC Parameters	15
4.3.1	Lifetime Configuration (“slow/standard” emitters): ≤ 20 MHz, ≥ 50 ns TAC Range	17
4.3.2	Lifetime Configuration (“fast” emitters): 80 MHz, 12.5 ns TAC Range	18
4.3.3	$g^{(2)}$ Configuration (“standard” emitters): 50 ns TAC Range	19
4.3.4	$g^{(2)}$ Configuration (“slow” emitters): 300 ns TAC Range	20
5	Optical Alignment Procedures	22
5.1	Beam Walking Alignment Procedure	22
5.2	Excitation Path	23
5.2.1	Acousto-Optic Modulator (AOM)	24
5.2.2	4f System: Spatial Filter & Telescope	27
5.3	Entering the Microscope: Main Dichroic Alignment	29
5.3.1	The “Quasi-Iris” Alignment Method	30
5.4	Confocal Collection and Objective Scanning	31
5.5	The Light-Tight Detector Box	32
5.5.1	Full Detector Box Alignment Procedure	33
5.5.2	Spectrometer Alignment	34
6	System Operation	34
6.1	Powering on devices	34
6.2	Loading Your Sample	35
6.3	Software Initialization	36
6.4	PL Scan Procedure	38
6.5	Spectrum Measurement	40

6.6	Stability & Saturation Measurements	41
6.7	Lifetime Measurement	42
6.8	$g^{(2)}(\tau)$ Photon Correlation Measurement	44
6.9	Automated Power Sweeping by Location (Batch Acquisition)	45
6.10	Autonomous Emitter Characterization (Autoscan)	46
6.10.1	Autoscan Configuration Procedure	47
6.10.2	Exiting Autoscan	48
7	Data Processing and Analysis	49
7.1	File Naming Convention and Metadata Extraction	49
7.2	Automated Metadata Parsing and Measurement Association	50
7.3	Interactive Data Exploration (<code>app.py</code>)	50
7.4	Automated Report Generation (<code>generate_report.py</code>)	51
7.5	Software Structure	52
8	Troubleshooting	53
	Appendix: Acronym Definitions	53

1 Introduction

This system is a fluorescence confocal time-correlated single-photon counting (TCSPC) system designed for characterizing quantum emitters. It can produce photoluminescence images, measure spectra and fluorescence decay, and perform second-order autocorrelation using a Hanbury Brown-Twiss (HBT) setup. It can operate autonomously or through a user-friendly interface equipped with features for an automated data collection and analysis pipeline. In this manual, I will introduce TCSPC and confocal microscopy to explain the optical and electronic design of the system. I will also provide chapters on how to operate and maintain the instrument. But first, here are the rules for this setup:

1. In operating this setup, you are using high-power lasers that can cause permanent harm to you (specifically your eyes). Always wear the appropriate laser safety goggles (rated for the specific operating wavelengths, typically OD6+ at 532 nm and 515 nm, as well as the IR range for the Mai Tai) when the beam path is uncontained.
2. You should never bring your face (and specifically your eyes) anywhere near the horizontal plane of the optical table. If you drop something on the floor, your natural reflex will be to bend down and pick it up immediately- train yourself not to do that. If you must make a movement that causes your eyes to cross the level of the table for any reason, make sure to close your eyes while making that movement.
3. Remove all reflective jewelry, watches, etc. and be mindful of where the beam is going; if you see an unaccounted reflection, bring it to someone's attention. We should never have laser beams traveling across the room.

Less important, but you can also damage the hardware if you are not careful:

4. The single-photon detectors can be destroyed by high-intensity light; never send the white light source or laser light directly into them.
5. The microscope objective is mounted directly onto the piezo stage in this setup. While the stage is rated to handle this load, it is not ideal for the piezo actuators to set the scanning speeds too high or to apply anything except gentle force to the objective. When you are done using the setup, it is best practice to turn the piezo controller off.
6. As detailed in Section 6.2, learn to use the torque adjustment and focus stopper rings of the microscope. If you feel mechanical resistance when turning the focus knobs, do not force it.

In general, if you're unsure about something, just ask. It's not a problem.

2 System Overview

The optical and electrical layouts of the system are presented in Fig. 1 and Fig. 2, with the component details listed in Table 1. The optical system is built around a modified Nikon Ti-U inverted microscope. Multiple continuous-wave and pulsed laser sources are co-aligned to the excitation path. A $100\times/0.90$ NA objective focuses the excitation laser and collects the resulting photoluminescence (PL). This objective is mounted on a three-axis piezo stage driven by a controller with analog input capabilities, allowing for precise positioning and raster scanning of the beam across a stationary sample at a 50 kHz sampling rate. A longpass dichroic filter reflects the excitation beam to the objective while transmitting the PL to an external tube lens and confocal pinhole; the light passing through the pinhole is then re-imaged onto the detectors.

The detectors are housed in a light-tight enclosure, accessible only through the confocal pinhole. Inside this enclosure are three Single-Photon Avalanche Diodes (SPADs) and a spectrometer. One SPAD is optimized for photon-counting and scanning, while the other two are optimized for time-correlated measurements. The system is electronically reconfigurable and optimized for various time-correlated measurements, such as lifetime and $g^{(2)}$. The TCSPC module and reconfigurable electronic layout are discussed in detail in Section 4. The excitation and PL paths, along with their alignment procedures, are discussed in detail in Section 5.

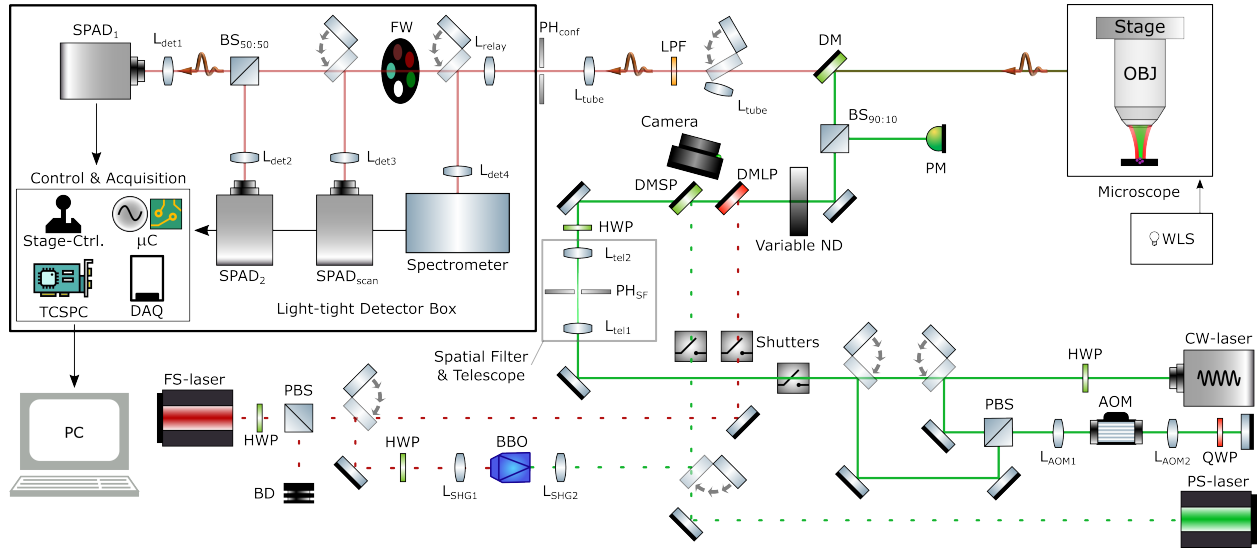


Figure 1: Layout of the optical setup.

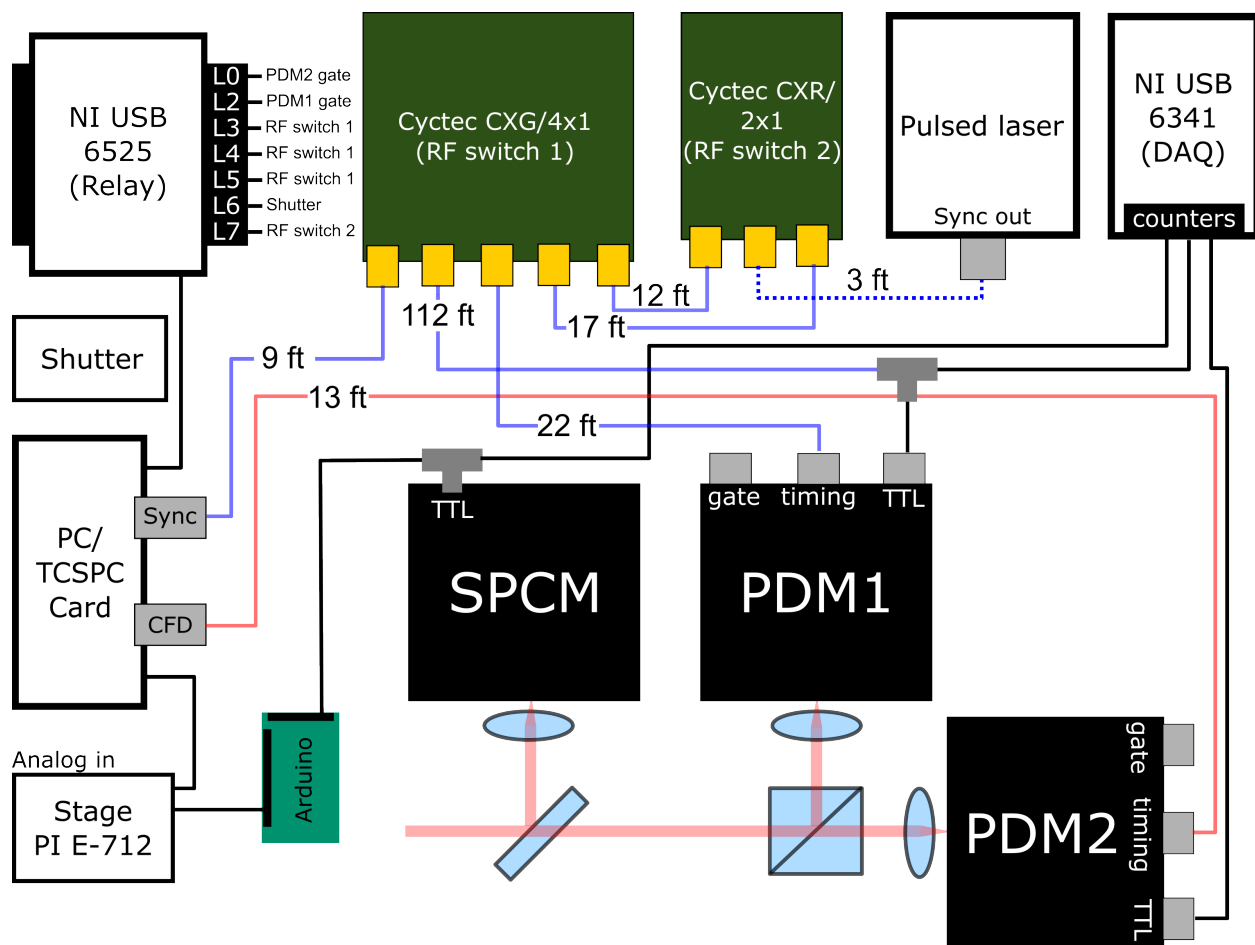


Figure 2: Layout of the electrical setup.

Table 1: Optical and electronic components.

Category	Label	Description / Manufacturer
Excitation Sources		
CW Laser	CW-laser	RGB Photonics (532 nm)
Pulsed Diode	PS-laser	Becker & Hickl (515 nm, variable rep rate (2, 20, 80 MHz), 40 ps pulse width)
Ultrafast Laser	FS-laser	Spectra-Physics Mai Tai HP (Ti:Sapphire, 80 MHz rep rate, 150 fs pulse width)
Nonlinear Crystal	BBO	Beta Barium Borate for Second Harmonic Generation
Lenses & Beam Shaping		
Spatial Filter	L_{tel1}, L_{tel2}	$f = 75$ mm and $f = 100$ mm ($M = 1.33$ Expansion)
AOM Relay	L_{AOM1}, L_{AOM2}	$f = 100$ mm (Acousto-optic relay)
SHG Control	L_{SHG1}, L_{SHG2}	$f = 25$ mm and $f = 50$ mm (Focusing/Collimation)
Confocal/Relay	L_{tube}, L_{relay}	Tube lens ($f = 200$ mm) and Detection relay ($f = 200$ mm)
Detection Focus	L_{det1-4}	$f = 45$ mm lenses for SPAD/Spectrometer coupling
Beam Control & Filtering		
Modulator	AOM	Gooch & Housego 3350-199 (Acousto-Optical)
Polarization	HWP, QWP	Half-Wave Plates and Quarter-Wave Plates
Beam Splitters	BS, PBS	Non-polarizing and Polarizing Beam Splitters
Dichroic Mirrors	DM, DMLP, DMSP	Main Dichroic, Long-Pass, and Short-Pass mirrors
Spectral Filters	LPF, FW	532 nm Long-Pass and Motorized Filter Wheel
Pinholes	PH_{SF}, PH_{conf}	25 μ m (Spatial Filter) and 100 μ m (Confocal)
Detection & Motion		
Stage & Objective	Stage, OBJ	PI-Mars Piezo Stage and Nikon 100 $\times$ Objective
Detectors	$SPAD_{1,2,scan}$	Single-Photon Avalanche Diodes
Spectroscopy	Spectrometer	Ocean Insight QE65000
Electronics	μ C, SC, TCSPC	Microcontroller, Stage Controller, and TCSPC Module

2.1 Detectors

It was mentioned in the previous section that one SPAD was optimized for photon-counting and scanning. This document will refer to this detector as the “SPCM” (Single-Photon Counting Mod-

ule). The other two detectors, optimized for timing, are referred to as “PDM1” and “PDM2” (Photon Detection Modules). Photon counting is done with the Transistor-Transistor Logic (TTL) outputs of the three SPADs wired to the counting channels of the data acquisition (DAQ) device (NI USB-6341). Time-correlated measurements are accomplished by routing the Nuclear Instrumentation Module (NIM) outputs of the laser and the two PDMs into the Becker & Hickl (B&H) TCSPC module. The SPCM, an Excelitas SPCM-AQRH-45, is the primary detector for general PL mapping and high-sensitivity measurements because it has good efficiency at the detection wavelengths of interest, and the timing resolution is suitable. The PDMs (MPD PDM Series) are reserved for ultrafast lifetime measurements because, despite their slightly lower efficiency, they have less timing jitter (< 50 ps FWHM). They utilize a smaller $50\text{ }\mu\text{m}$ custom epitaxial silicon SPAD to minimize transit time variations and also provide decoupled outputs: a standard TTL pulse for rate monitoring and a fast NIM pulse specifically for high-resolution timing. The specifications for these two detectors are listed below in Table 2.

Table 2: Comparison of Detector Characteristics

Characteristic	Excelitas SPCM-AQRH-45	MPD PDM
Usage	PL mapping and counting.	Time-resolved measurements.
Efficiency (PDE)	75 % @ 650 nm 50 % @ 830 nm	50 % @ 550 nm 37 % @ 650 nm 12 % @ 830 nm
Active Area	$\varnothing 180\text{ }\mu\text{m}$	$\varnothing 50\text{ }\mu\text{m}$
Timing Jitter	225 ps FWHM (typ.)	< 50 ps FWHM (typ.)
Dead Time	22 ns	77 ns
Dark Count Rate	100 s^{-1}	$< 250\text{ s}^{-1}$ (typical)
Outputs	TTL	TTL 250 ps FWHM NIM 35 ps FWHM

Timing jitter, or transit time spread (TTS), is an important parameter for SPADs. TTS represents the statistical uncertainty of the detection event. This arises from stochastic fluctuations in the delay between photon absorption and the onset of the electrical avalanche. In a histogram of photon arrival times, this jitter manifests as the broadening of the instrument response function (IRF), quantified by the Full Width at Half Maximum (FWHM). The total system jitter is the root sum square of the uncertainties from the pulsed laser, the detector’s internal jitter, and the timing electronics. Measuring the IRF will be discussed in more detail in Section 4.

To further elaborate on the differences between the two SPADs, we should understand their outputs. As mentioned, they both output TTL, but the PDMs are capable of outputting NIM pulses. TTL is a robust, voltage-based standard ($> 2.2\text{ V}$). Due to relatively slow rise times ($\approx 1\text{--}2\text{ ns}$), TTL is used here for the SPCM and the PDM rate monitoring channels. NIM is a fast, current-based standard ($\approx -800\text{ mV}$ into $50\text{ }\Omega$). The MPD NIM output features an extremely sharp 450 ps falling edge, which is essential for achieving sub-ns timing resolution.

The relevant pulses from the detectors and lasers relevant for the TCSPC have been measured, as shown in Fig. 3. They will be used in Section 4.3 to set the optimal TCSPC parameters.

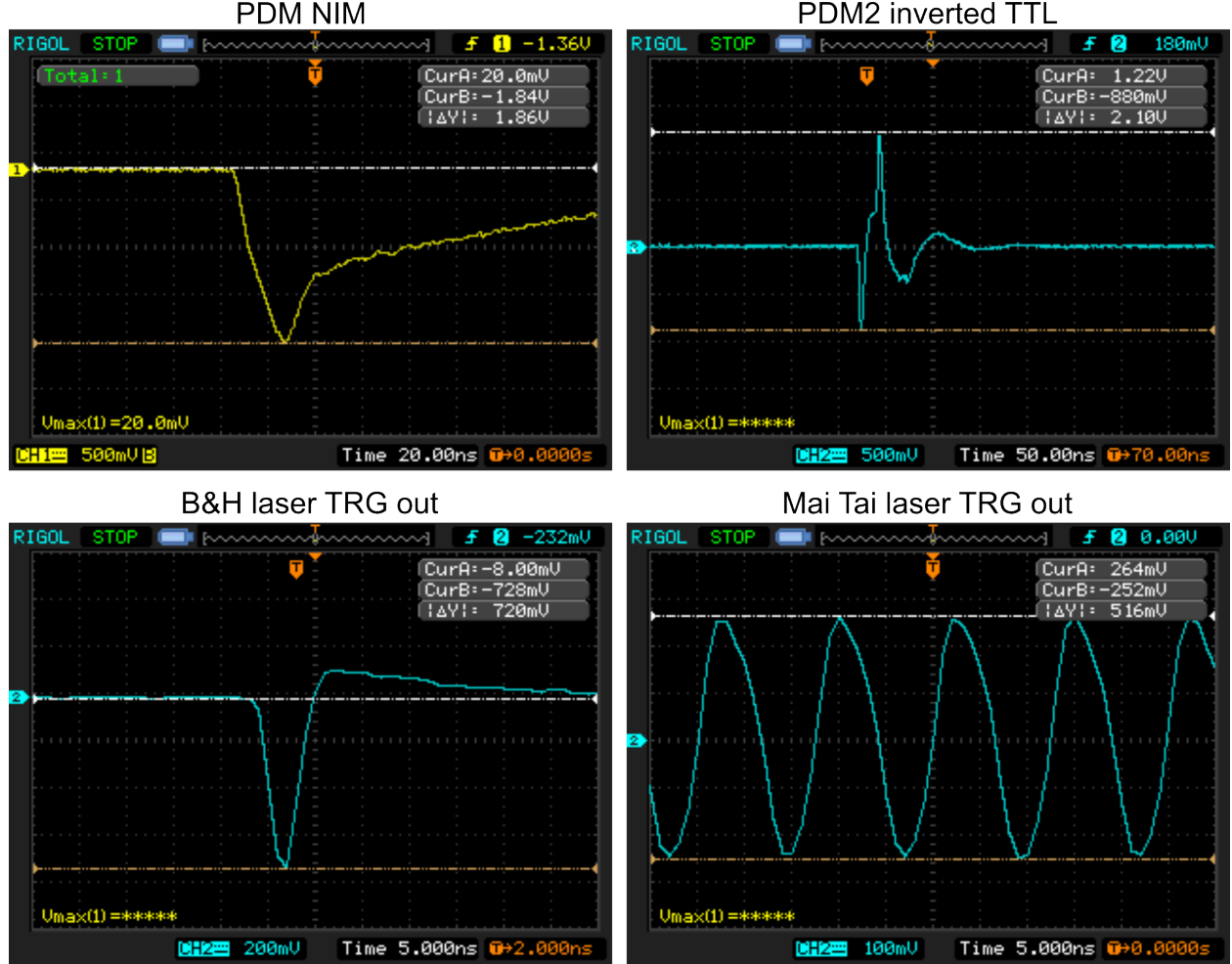


Figure 3: Oscilloscope comparison of detector and laser outputs.

Note that the temporal profiles shown in Fig. 3 are bandwidth-limited by the oscilloscope. Because the intrinsic widths of these fast pulses approach or exceed the analog bandwidth of the measurement instrument, the displayed waveforms exhibit artificial broadening and amplitude attenuation. Therefore, these traces serve to verify signal presence, polarity, and general macroscopic shape rather than absolute temporal characteristics. The PDM2 pulse (Top Right) exhibits a distorted shape because it is not a native NIM pulse; rather, it is a TTL pulse converted to a NIM-compatible standard via an in-line pulse inverter (B&H A-PPI-D) and a 10 dB SMA attenuator. Similarly, while the Mai Tai laser output does not adhere to the NIM protocol, it can still function reliably as the TCSPC SYNC signal, provided the constant fraction discriminator (CFD) parameters are properly optimized, as will be detailed in Section 4.3.

2.2 Photon-counting

The custom LabVIEW interface provides real-time monitoring of the photon counts across all three detectors. Pulse counting is handled by the NI USB-6341 DAQ device, utilizing three Programmable Function Interface (PFI) terminals configured as counter inputs. The TTL output from each SPAD is routed to an independent counter channel, with the specific pinout detailed in Table 3.

Table 3: Counter Signal Routing and PFI Assignments

Signal Function	Counter 0 - SPCM	Counter 1 - PDM1	Counter 2 - PDM2
SOURCE	PFI 8 (Pin 81)	PFI 3 (Pin 76)	PFI 0 (Pin 73)
GATE	-	USB-6525 (L2)	USB-6525 (L1)

The NIM and TTL outputs from all three detectors are designed to drive a $50\,\Omega$ load. The coaxial cables transmitting these signals function as transmission lines with a characteristic impedance of $Z_0 = 50\,\Omega$. To prevent signal reflection, a TTL pulse traveling down this line must be properly absorbed at the receiver. However, the PFI terminals on the DAQ exhibit a relatively high input impedance (approximately $50\,\text{k}\Omega$) compared to the $50\,\Omega$ source impedance of the SPADs. This severe impedance mismatch causes the majority of the pulse energy to reflect at the DAQ interface. The reflected wave then propagates back and forth along the cable, inducing voltage doubling and capacitive ringing on the signal’s rising edge.

To resolve this, $50\,\Omega$ parallel termination resistors were installed across the respective PFI and ground terminals at the DAQ inputs. This termination matches the characteristic impedance of the transmission line, safely dissipating the TTL pulse energy and suppressing the ringing that could otherwise cross the DAQ’s logic thresholds and trigger false photon counts. While the active quenching circuitry and inherent dead time of the SPADs mitigate most of these reflection effects, implementing the termination circuit is best practice for ensuring signal integrity.

2.3 High-speed Analog Scanning

The system acquires PL images by raster scanning the objective point-by-point, translating the focused laser beam across the stationary sample and recording the photon counts at each location. A conventional “step-settle-record” protocol would be inefficient due to USB communication latency and stage settling times. To achieve fast imaging rates, the system operates in a continuous scanning mode.

In this configuration, the piezoelectric stage moves the objective in a continuous waveform (e.g., a sine or ramp wave) while the E-712 piezo controller acts as a synchronized DAQ unit. Instead of reading the photon counter via USB at every pixel, the photon count rate is converted into a proportional analog voltage by a microcontroller (Arduino Mega) and fed directly into the E-711.IA4 analog interface module installed in the E-712 controller. The E-712 then simultaneously records the exact spatial position and light intensity in its internal memory at the servo update rate

(up to 50 kHz). The resulting data array is subsequently processed by a MATLAB script integrated within the main LabVIEW program to reconstruct the two-dimensional PL scan.

Table 4: Arduino wiring. The stage is connected to using the "Analog In" port on its controller. The SPCM is connected from it's TTL output. Be sure to connect pin A7 to D2 with a jumper wire.

Signal	Connection Point	Arduino Pin / Reference
Stage Positive	DAC output	DAC0
Stage Negative	Ground	GND
SPCM Positive	Analog input	A5
SPCM Negative	Ground	GND
Arduino Internal	–	A7 → D2

3 Confocal Microscopy

Confocal microscopy is an optical imaging technique that provides a significant advantage over conventional wide-field microscopy by utilizing a spatial pinhole to reject out-of-focus light and stray light thereby greatly increasing the signal-to-noise ratio. The trade-off is that the image is collected point-by-point. By raster-scanning a focused excitation laser across the sample and collecting fluorescence through this conjugate pinhole, the system creates sharp, high-resolution optical sections. This capability is important for studying isolated quantum emitters, where background fluorescence from the bulk substrate must be minimized. While this technique allows for a volumetric image of a bulk sample to be scanned, for our purposes, we typically just image the surface plane of our sample, as we mostly work with flat devices.

3.1 Infinity-Corrected Optics

Our microscope objective (Nikon LU Plan 100× Objective) is infinity-corrected, meaning it projects light from the sample as a collimated beam (i.e., parallel rays) rather than directly forming an image. The region containing this collimated light is known as the infinity space. The presence of the infinity space offers several advantages, the most important being the availability of additional space in which optical components can be inserted without significantly affecting image formation.

To form a magnified image, a second lens, known as the tube lens, is placed in the optical path to refocus the collimated light. The total magnification (M) of the system is determined by the ratio of the tube lens focal length (f_{TL}) to the objective focal length (f_{obj}):

$$M = \frac{f_{TL}}{f_{obj}} \quad (1)$$

For standard Nikon infinity-corrected microscope systems, including this one, the tube lens has a fixed focal length of $f_{TL} = 200$ mm. In the Nikon Ti-U microscope, the internal optics can direct the image either to the eyepieces or to the Right Side Port (RSP). Instructions for microscope

operation are in section 6.2. The eyepiece optical path contains a built-in tube lens that forms a real intermediate image for direct viewing. Under normal operation, the RSP also includes an internal tube lens that forms an image on a camera sensor; however, in our system, this tube lens has been removed.

With the internal tube lens removed from the RSP, the fluorescence signal exits the microscope body as a collimated beam in the infinity space. This configuration allows us to insert a 550 nm long-pass dichroic mirror and additional optical elements into the collimated beam path. Operating in the infinity space provides extra space for optics and minimizes angle-dependent shifts and potential distortions that can occur when components are placed in a converging or diverging beam.

3.2 Simulation of Confocal Principle

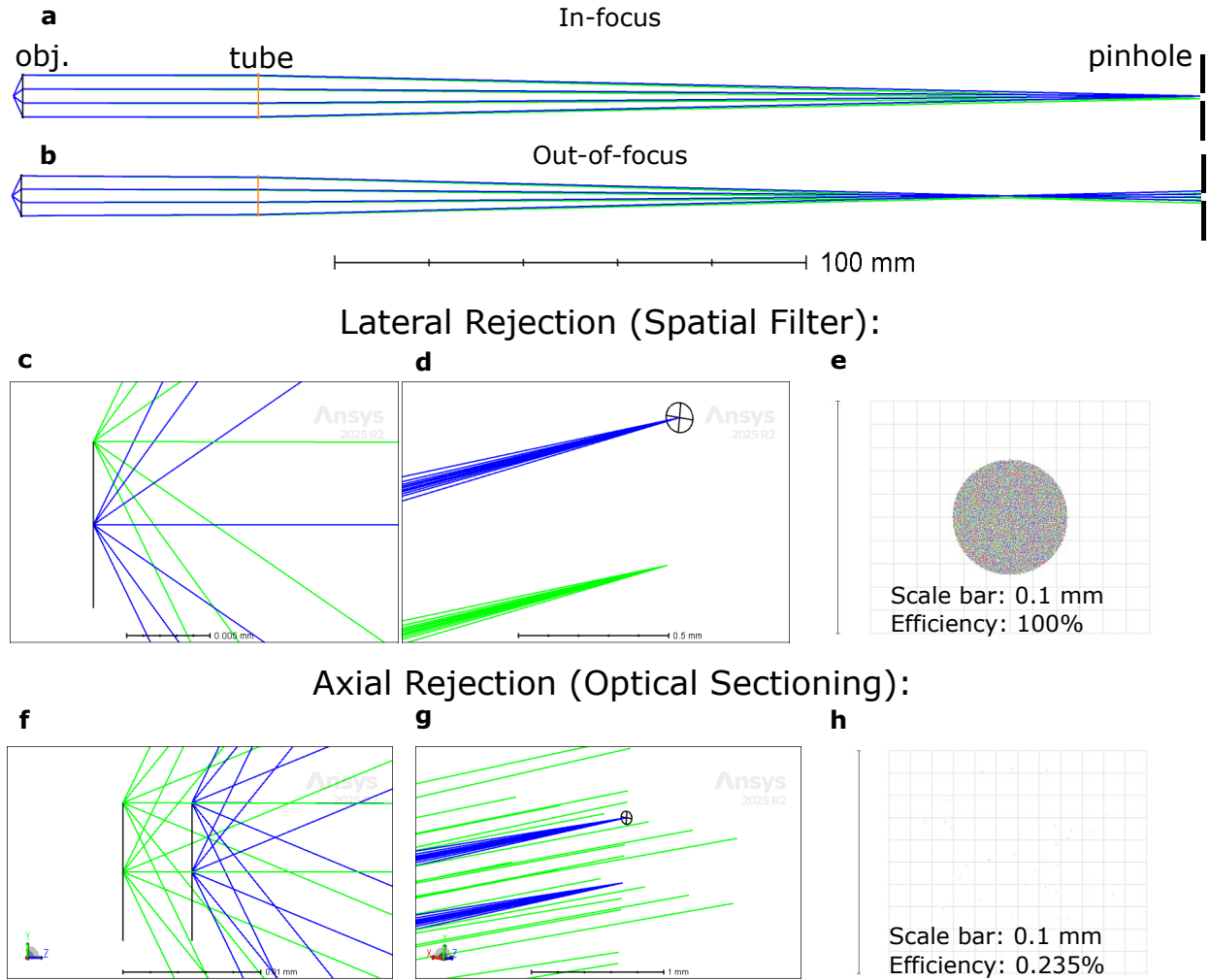


Figure 4: Zemax simulation of a simple confocal microscope. The light originating from the focal plane passes through the conjugate pinhole, while the off-axis or out-of-focus light sources are blocked by the pinhole.

Fig. 4 presents a Zemax OpticStudio simulation demonstrating the principles of confocal microscopy. Fig. 4A models the optical system using paraxial lenses for the objective ($f = 2\text{ mm}$) and tube lens ($f = 200\text{ mm}$), achieving a $100\times$ magnification. The infinity space is shown between the objective and tube lens, which here is 50 mm but in the actual setup spans over a meter and contains the dichroic mirror and emission filters. The system is imaged onto a $100\text{ }\mu\text{m}$ aperture, representing the confocal pinhole at the simulation’s image conjugate plane. Light originating precisely from the focal plane of the objective is tightly focused and passes through the pinhole (Fig. 4A). Conversely, light from out-of-focus planes expands at the pinhole plane; the majority of this defocused light is physically blocked by the aperture walls. This is illustrated in Fig. 4B, where the simulated field points are displaced $5\text{ }\mu\text{m}$ along the optical axis.

Fig. 4C–E illustrate the lateral rejection of light, effectively acting as a spatial filter. As shown in Fig. 4C, two field points are positioned in the focal plane: one on-axis and one displaced $5\text{ }\mu\text{m}$ off-axis in the Y-direction. Ray tracing to the image plane (Fig. 4D) reveals that the on-axis point transmits 100% of its energy through the pinhole, which is confirmed by the geometric image analysis in Fig. 4E. In contrast, the off-axis point is imaged $500\text{ }\mu\text{m}$ away from the center, missing the pinhole entirely.

Finally, Fig. 4F–H demonstrate the axial rejection of out-of-focus light. This simulation models four distinct field points (Fig. 4F): an on-axis and a $5\text{ }\mu\text{m}$ off-axis point at the focal plane, paired with an identical set situated $5\text{ }\mu\text{m}$ out-of-focus. As shown in Fig. 4G, the out-of-focus rays are heavily defocused at the confocal plane, while the off-axis point remains completely rejected as in the previous example. Only a negligible fraction of the energy from the out-of-focus plane enters the pinhole; the geometric image analysis in Fig. 4H calculates this transmission to be only 0.235% .

4 Time-Correlated Single-Photon Counting (TCSPC)

TCSPC is a high-precision statistical technique used to measure fluorescence lifetimes and photon correlation statistics. TCSPC detects individual photons and times their arrival relative to another pulse, whether that be a pulsed excitation source (laser) or another photon detection event. Over millions of excitation cycles, a histogram of these arrival times is built, reconstructing the photon arrival-time data with picosecond-level temporal resolution. In this chapter, I overview the working principles of the TCSPC module using information from the B&H TCSPC Handbook [1] (which is an incredible useful resource if you are interested in learning more) and the specific electronic layout of our system.

4.1 Working Principle

The core of the timing system is the TCSPC module (B&H SPC-150N), which is installed in the setup’s PC via a direct motherboard connection. One might wonder what advantages this module offers over a standard high-speed oscilloscope. A typical oscilloscope would be overwhelmed by the sustained data rates involved in single-photon counting, and recording continuous time traces

would quickly consume prohibitive amounts of storage, making it impractical for these experiments. The TCSPC module is a specialized circuit designed to measure the time delay between two signals with low jitter and minimal dead time; it is incredibly accurate and rarely misses the fast NIM pulses our SPADs output. Furthermore, it eliminates the data storage bottleneck by processing and binning the arrival times into a histogram directly on the onboard hardware. Fig. 5 shows the high-level layout of the B&H SPC-150N TCSPC module.

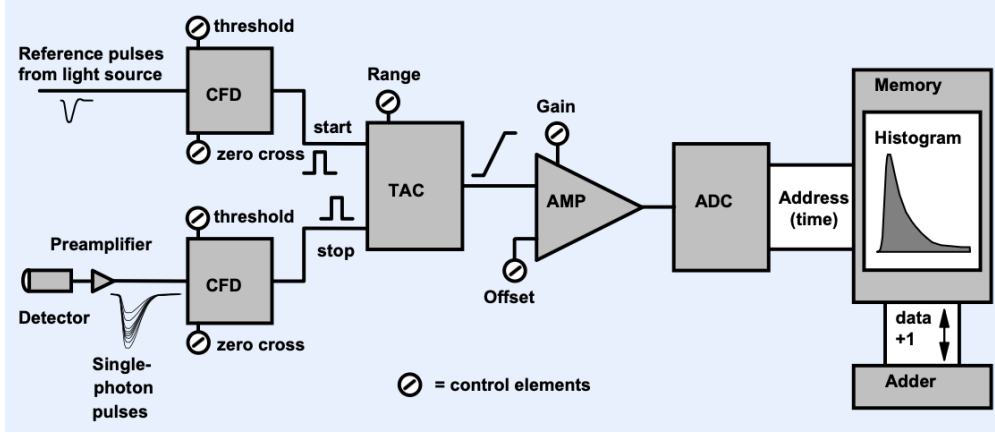


Figure 5: TCSPC module, reproduced with permission from Becker & Hickl.

The detection electronics must handle pulses from single-photon detectors, which often exhibit considerable amplitude instability. A simple Leading-Edge Discriminator would translate this amplitude stability into timing jitter. To mitigate this, the system uses a constant fraction discriminator (CFD). The CFD triggers at a constant fraction of the pulse amplitude rather than a fixed voltage threshold, effectively decoupling the timing precision from the pulse amplitude variations. The triggering of the CFDs is used to start and stop the time-to-amplitude converter (TAC). The TAC generates an output signal proportional to the time delay between the start and the stop pulse. Conventional TACs use a switched current source charging a capacitor. While B&H does not publish the exact proprietary schematic of their TAC, it operates on this same fundamental principle. The start pulse switches the current on, and the stop pulse turns it off. If the current in the start-stop interval is constant, the final voltage at the capacitor represents the time between start and stop. This principle works with remarkable accuracy, and time differences of a few picoseconds (ps) can be clearly resolved. The TAC output voltage is sent through a biased amplifier (AMP). The amplifier has a variable gain and a variable offset. The offset is used to select a smaller time window within the full-scale conversion range of the TAC. The amplified TAC signal is fed to the analog-to-digital converter (ADC). The output of the ADC is the digital equivalent of the photon detection time.

There are two CFDs; one provides the start signal for the TAC, and the other provides the stop. They have SMA input connections and are labeled Sync (start) and CFD (stop) on the physical TCSPC module. Their specific wiring depends on the measurement goals- such as whether acquiring fluorescence lifetimes or $g^{(2)}(\tau)$ autocorrelation- and the relevant time scales. This will be detailed

in Section 4.3.

4.2 Reverse Start-Stop Configuration

It is important to note that the system actually operates in what is called a “Reverse Start-Stop” configuration. The system typically operates with a MHz laser repetition rate, while the photon count rate at the detector is often in the kHz to low MHz range (due to quantum yield and collection efficiency limits). If operated in a standard “Forward” mode (where the laser starts the TAC and a detected photon stops it), the TAC would be triggered millions of times per second but only successfully stopped a small fraction of the time. This would force the TAC to reset continuously, resulting in excessive dead time.

To solve this, the system is operated in Reverse Start-Stop mode: the TAC is started by the detection of a photon (CFD Detector) and stopped by the subsequent laser pulse (CFD Sync). This ensures the TAC only cycles when a valid photon is detected. However, because the photon is measured relative to the *next* laser pulse rather than the one that generated it, this introduces a new problem: the timing jitter between sequential laser pulses is incorporated into the measurement. Fortunately, there is a simple hardware solution. The length of the laser Sync cable can be increased so that it acts as a passive delay line, physically delaying the signal’s arrival at the Sync CFD by approximately one pulse period.

4.3 Delay Line Configurations and Optimal TCSPC Parameters

As a quick rule of thumb, light travels about one foot per nanosecond. Signals in our coaxial cables propagate at roughly $0.66c$. Cable length isn’t the only source of delay; we also have to account for detector response times, RF switch propagation time, internal routing within the TCSPC module itself, and other factors we cannot easily predict. Because of these extra variables, the required delay lengths are estimated upfront and then tuned experimentally. We physically swap in different lengths of cable until the decay curves and $g^{(2)}$ dips are properly centered in the measurement window. For this setup, the CFD cable length is kept locked at 13 ft, leaving the SYNC cable as our variable adjustment. To accommodate the timing requirements of both lifetime spectroscopy and photon correlation ($g^{(2)}(\tau)$) measurements across varying timescales, the system features an electronically switchable delay line architecture. This allows the user to switch between measurement modes and timescales via software without physically recabling the detector inputs. Table 5 summarizes the final, optimized cable lengths for the four main measurement settings.

Table 5: Signal routing and delay line configurations. Delay lengths are representative values required to center the photon distribution within the TAC window.

Mode	Window	Start	Stop	CFD length (ft)	SYNC length (ft)
Lifetime (80 MHz)	12.5 ns	PDM1	Laser Sync	13	29
Lifetime (20 MHz)	50 ns	PDM1	Laser Sync	13	24
$g^{(2)}(\tau)$ (Short)	50 ns	PDM1	PDM2	13	31
$g^{(2)}(\tau)$ (Long)	300 ns	PDM1	PDM2	13	121

Note in case of disaster: In case the wiring needs to be redone, here are the exact combinations of wire lengths I used to get the correct lengths: 29 ft **80 MHz LT wire** $\rightarrow$ 9 ft from PC to RF1.0 (6 ft + 3 ft), 12 ft from RF1.4 to RF2.0 (6 ft + 6 ft), 3 ft from RF2.1 to laser. 24 ft **20 MHz LT wire** $\rightarrow$ 9 ft from PC to RF1.0 (6 ft + 3 ft), 17 ft from RF1.3 to RF2.2 (6 ft + 6 ft + 150 cm), 3 ft from RF2.1 to laser. 31 ft **50 ns $g^{(2)}$ wire** $\rightarrow$ 9 ft from PC to RF1.0 (6 ft + 3 ft), 22 ft from RF1.2 to PDM2 (300 cm + 6 ft + 6 ft). 121 ft **300 ns $g^{(2)}$ wire** $\rightarrow$ 9 ft from PC to RF1.0 (6 ft + 3 ft), 112 ft from RF1.1 to PDM1 TTL (640 cm + 1280 cm + 1280 cm); a pulse inverter and 10 dB SMA attenuator are needed in this line (see Section 4.3.4).

The routing is managed by a set of RF switches (Cytec CX-series) controlled by a **National Instruments USB-6525** digital relay device. LabVIEW communicates with this interface to activate specific delay line configurations, ensuring that the “Stop” pulse always arrives within the linear conversion range of the TAC relative to the “Start” pulse. Table 6 is included in case the LabVIEW programming for the relay or the hardware wiring to the RF switches needs to be reconstructed.

Table 6: Hardware Connections and Measurement Settings for NI Relay USB-6525

Software Line	Hardware Line	LabVIEW var	LabVIEW write (T/F)				Connected To
			S-LT	L-LT	S- $g^{(2)}$	L- $g^{(2)}$	
Dev2/port0/line0	P0.0 A/B	PDM 2 gate	F	F	F	F	PDM2 gate
Dev2/port0/line2	P0.2 A/B	PDM 1 gate	F	F	F	F	PDM1 gate
Dev2/port0/line3	P0.3 A/B	setting 1	F	T	F	F	Cytec 4x1.1
Dev2/port0/line4	P0.4 A/B	setting 2	F	F	T	F	Cytec 4x1.2
Dev2/port0/line5	P0.5 A/B	setting 3	F	F	F	T	Cytec 4x1.3
Dev2/port0/line6	P0.6 A/B	pulse laser on/off	F	F	F	F	Shutter (EOPC)
Dev2/port0/line7	P0.7 A/B	setting 4	T	F	F	F	Cytec 2x1

Note: S-LT = Short Lifetime, L-LT = Long Lifetime, S- $g^{(2)}$ = Short $g^{(2)}$, L- $g^{(2)}$ = Long $g^{(2)}$.

Note in case of disaster: In case the soldering to the RF switches comes undone and a quick-fix is needed, here are the ribbon cable mappings: **Cytec 4x1.1** $\rightarrow$ green/blue (2nd from top), **Cytec 4x1.2** $\rightarrow$ purple/grey (2nd from top), **Cytec 4x1.3** $\rightarrow$ purple/grey (1st from top), and **Cytec 2x1** $\rightarrow$ orange/yellow.

4.3.1 Lifetime Configuration (“slow/standard” emitters): ≤ 20 MHz, ≥ 50 ns TAC Range

As discussed in the previous section, there are currently four different measurement mode timescales pre-programmed into the LabVIEW software: two lifetime configurations and two $g^{(2)}$ configurations, each optimized for a specific temporal regime.

This should be the default setting for measuring lifetimes unless the emitter decays fast enough that the curve can fit within a 12.5 ns TAC range window (in which case, the configuration in the next section should be used). For this measurement, the full delay line architecture (Fig. 2) is implemented as shown in Fig. 6:

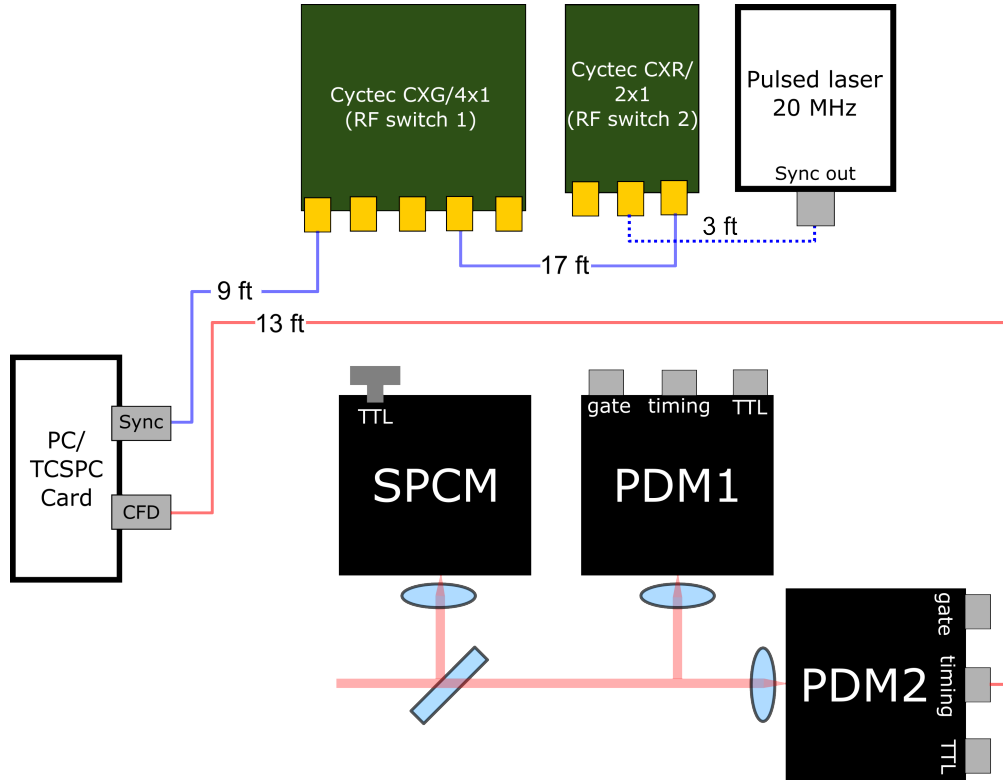


Figure 6: Lifetime configuration for a laser rep rate of 20 MHz (or less). TAC range set to 50 ns.

The delay lines for this configuration allow for a 50 ns TAC range; consequently, the maximum repetition rate that can be used is 20 MHz ($T = 1/f = 1/(20 \times 10^6) = 50$ ns). A 2 MHz repetition rate with a manual override of the TAC range to 500 ns can also work with this routing if needed to measure an emitter with a very long (hundreds of nanoseconds) lifetime. You will need to change the TAC range after selecting the correlation measurement (shown in 20) because using that selector button auto populates the settings. If you are doing something like this, it’s recommended to check the B&H handbook[1]. The reason for the selection of 80, 20, and 2 MHz is because the Becker & Hickl laser only supports these repetition rates. Note that the delay line is not long enough to lock the photon-creating pulse to the measured CFD pulse (the technique discussed in Section 4.2

to eliminate laser jitter). This is intentionally not implemented here because the laser jitter is on the picosecond scale, and this setting is intended for lifetimes of at least several nanoseconds.

The optimal TCSPC parameters for this configuration are shown below in Table 7:

Table 7: Optimal TCSPC Settings for 20 MHz Lifetime Measurement

Discriminator Settings	Start (CFD)	Stop (SYNC)
Threshold (LL)	−51 mV	−51 mV
Zero Cross (ZC)	−5.29 mV	−5.29 mV
TAC / Time Range Settings		
Range	50 ns	
Gain	1	
Offset	6.0000 %	
Limit High	97.000 %	
Limit Low	3.000 %	
Frequency Divider	1	

These parameters were derived from the pulse characterizations of the laser and detector shown in Fig. 3 and further optimized through experimental tuning. This statement applies to all following configurations as well.

4.3.2 Lifetime Configuration (“fast” emitters): 80 MHz, 12.5 ns TAC Range

This configuration should be used when measuring very fast (sub-nanosecond) decay times. It includes an extra 12–13 ft SYNC delay line length to successfully lock the detected CFD pulse to the exact laser pulse that created it, utilizing the technique defined in Section 4.2 to eliminate laser jitter from the measurement. For this measurement, the active delay line routing is configured as shown in Fig. 7:

In this mode, PDM2 provides the fixed “Start” signal, while PDM1 provides the “Stop” signal. Because the photon arrival times are random relative to one another, the signal from PDM1 must be delayed by half the TAC window width to shift the $t = 0$ coincidence peak to the center of the generated histogram.

For this measurement, the active delay line routing is configured as shown in Fig. 8:

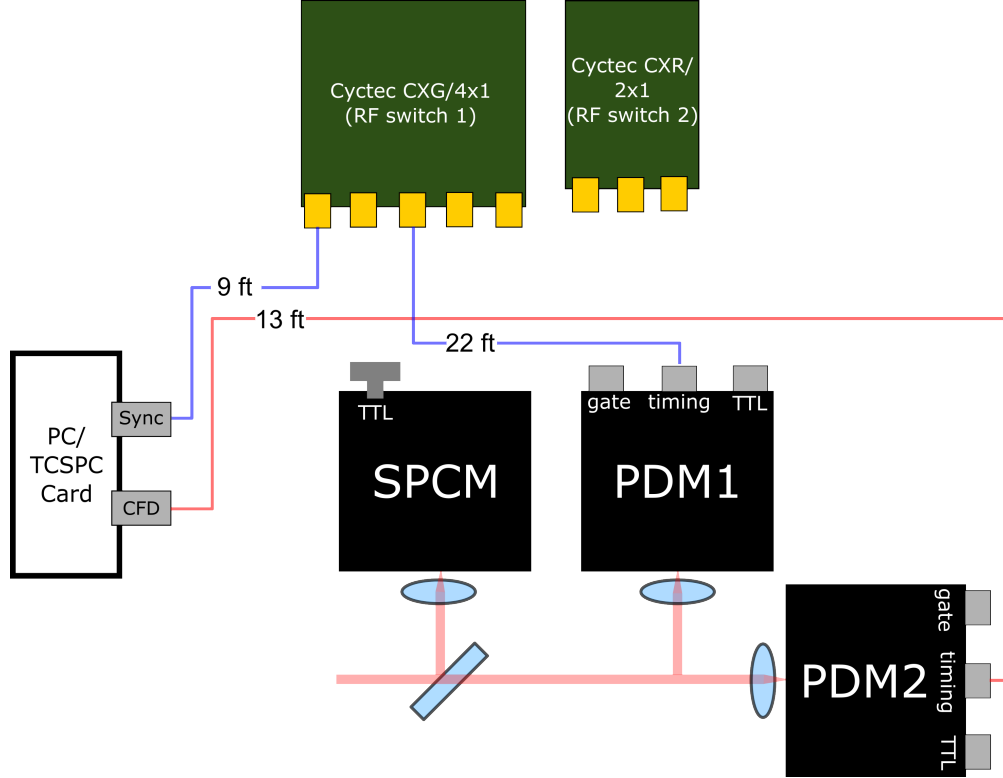


Figure 8: 50 ns TAC Range $g^{(2)}$ configuration.

This is the standard $g^{(2)}$ measurement configuration, as a 50 ns TAC range is appropriate for most emitters. The optimal TCSPC parameters are shown in Table 9:

4.3.4 $g^{(2)}$ Configuration (“slow” emitters): 300 ns TAC Range

This setting allows for a 300 ns TAC range window and requires 121 ft of delay line. For this measurement, the active delay line routing is configured as shown in Fig. 9:

Table 9: Optimal TCSPC Settings for 50 ns $g^{(2)}$ Measurement

Discriminator Settings	Start (CFD)	Stop (SYNC)
Threshold (LL)	-51 mV	-51 mV
Zero Cross (ZC)	-5.29 mV	-5.29 mV
TAC / Time Range Settings		
Range	50 ns	
Gain	1	
Offset	6.0000 %	
Limit High	97.000 %	
Limit Low	3.000 %	
Frequency Divider	1	

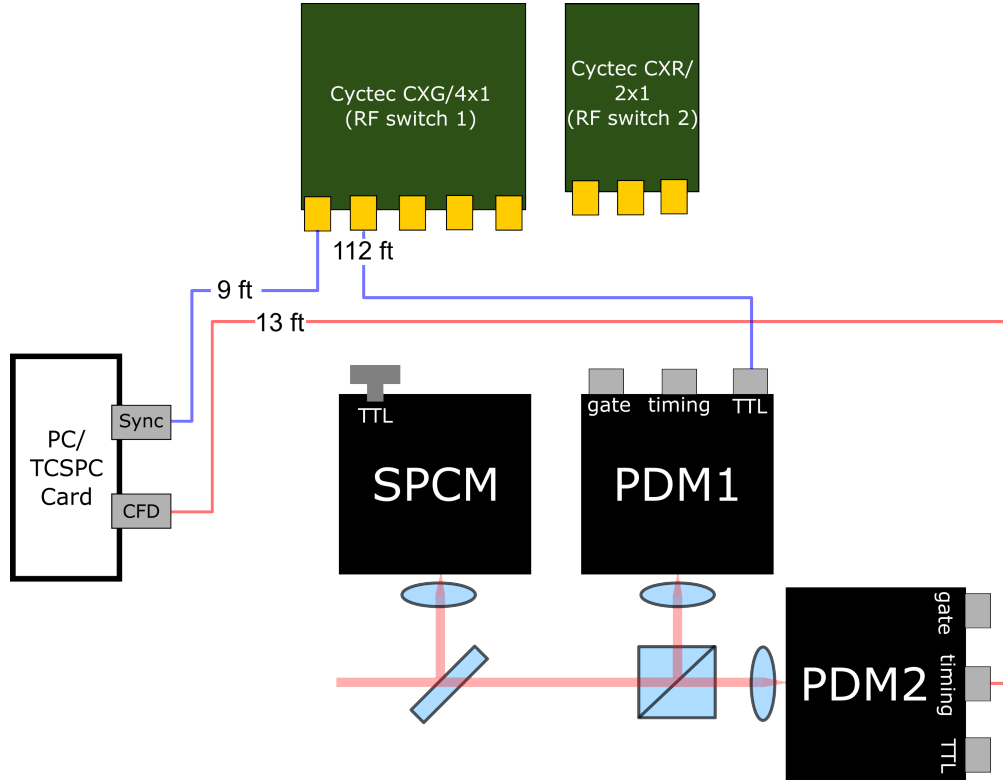


Figure 9: 300 ns TAC Range $g^{(2)}$ configuration. An inverter on the PDM1 TTL line is not pictured but is present in the setup.

This configuration should be used when measuring very wide $g^{(2)}$ windows, which is primarily necessary for emitters exhibiting long lifetimes or complex shelving dynamics. This specific routing line is the one exception to the rule of exclusively using NIM/timing outputs for the TCSPC measurements. To keep the hardware setup less complicated and avoid adding extra dedicated routing, the standard TTL output from PDM1 is "converted" into a NIM-compatible pulse using an in-line pulse inverter (B&H A-PPI-D) and a 10 dB SMA attenuator.

The optimal TCSPC parameters for this configuration are shown below in Table 10:

Table 10: Optimal TCSPC Settings for 300 ns $g^{(2)}$ Measurement

Discriminator Settings	Start (CFD)	Stop (SYNC)
Threshold (LL)	−51 mV	−51 mV
Zero Cross (ZC)	−5.29 mV	−5.29 mV
TAC / Time Range Settings		
Range	300 ns	
Gain	1	
Offset	10.0000 %	
Limit High	97.000 %	
Limit Low	3.000 %	
Frequency Divider	1	

5 Optical Alignment Procedures

Aligning a confocal microscope coupled to a TCSPC system requires a systematic approach and patience. The ultimate goal is to overlap the excitation spot (laser) and the collection spot (pinhole) at the sample plane with sub-micron accuracy. Since the system is pre-established, requiring a complete, from-scratch realignment should be a rare event. This guide breaks the optical path down into distinct sections that can be aligned independently; connecting them is then trivially achieved using steering mirrors.

The user should become familiar enough with the setup to recognize where a potential misalignment originates and know which specific section requires correction. For example, if the beam quality entering the microscope is poor, the issue is likely the upstream $4f$ spatial filter. If the photon counts are high on one PDM but low on the other, it likely requires a simple fine-tuning of the final lens or mirror immediately preceding the misaligned detector. If all detector counts are low, the confocal pinhole may be misaligned. It is important to note that final, fine-tune alignment should always be performed using a point-source object to ensure precision. I recommend using a nanodiamond nitrogen-vacancy (NV) center with benchmarked parameters (e.g., a known count rate at a specific excitation power). This will be detailed in Section 5.5.

5.1 Beam Walking Alignment Procedure

If you are unfamiliar with optical alignment, it is good practice to use a systematic approach rather than "guessing and checking." The standard method for aligning a beam to a defined optical axis (established by two irises or apertures) is called beam walking. To use this method, you need two kinematic mirrors (**M1**, **M2**) and two alignment targets/irises (**T1**, **T2**).

- **M1 (Upstream Mirror):** Controls the beam's spatial position at the first target (**T1**).

- **M2 (Downstream Mirror):** Controls the beam’s pointing angle toward the final target (T2).
- **T1 (Proximal Target):** An iris located immediately after M2.
- **T2 (Distal Target):** An iris located further downstream, such as at the entrance of the microscope objective.

Perform the following iterative cycle:

1. **Positioning (M1 $\rightarrow$ T1):** Close iris T1 to a small aperture. Adjust the kinematic knobs on M1 until the beam is perfectly centered on T1. Open T1.
2. **Pointing (M2 $\rightarrow$ T2):** Close iris T2 to a small aperture. Adjust the kinematic knobs on M2 until the beam is perfectly centered on T2.
3. **Iteration:** Adjusting M2 in Step 2 inevitably shifts the beam slightly off-center at T1. Therefore, repeat Steps 1 and 2 iteratively. With each cycle, the beam will walk closer to the true axis until it passes cleanly through the center of both T1 and T2 simultaneously.

Note: Convergence is fastest when the distance between M1 and M2 is maximized, and the distance between M2 and T1 is minimized. If the mirrors are too close together, the angular leverage of M1 is reduced, requiring more iterations to converge.

5.2 Excitation Path

The laser excitation path is shown in Fig. 10. The most notable sub-assemblies are the acousto-optic modulator (AOM), the $4f$ spatial filter system, and the dichroic mirrors. Generally speaking, the system is aligned by walking the beam through strategically placed irises. The goal is to have the laser(s) enter the microscope perfectly co-aligned, coincident to the optical axis, and centered on the back aperture of the objective. We must also verify that the beam maintains its Gaussian profile as it propagates through the objective without clipping on the entrance aperture.

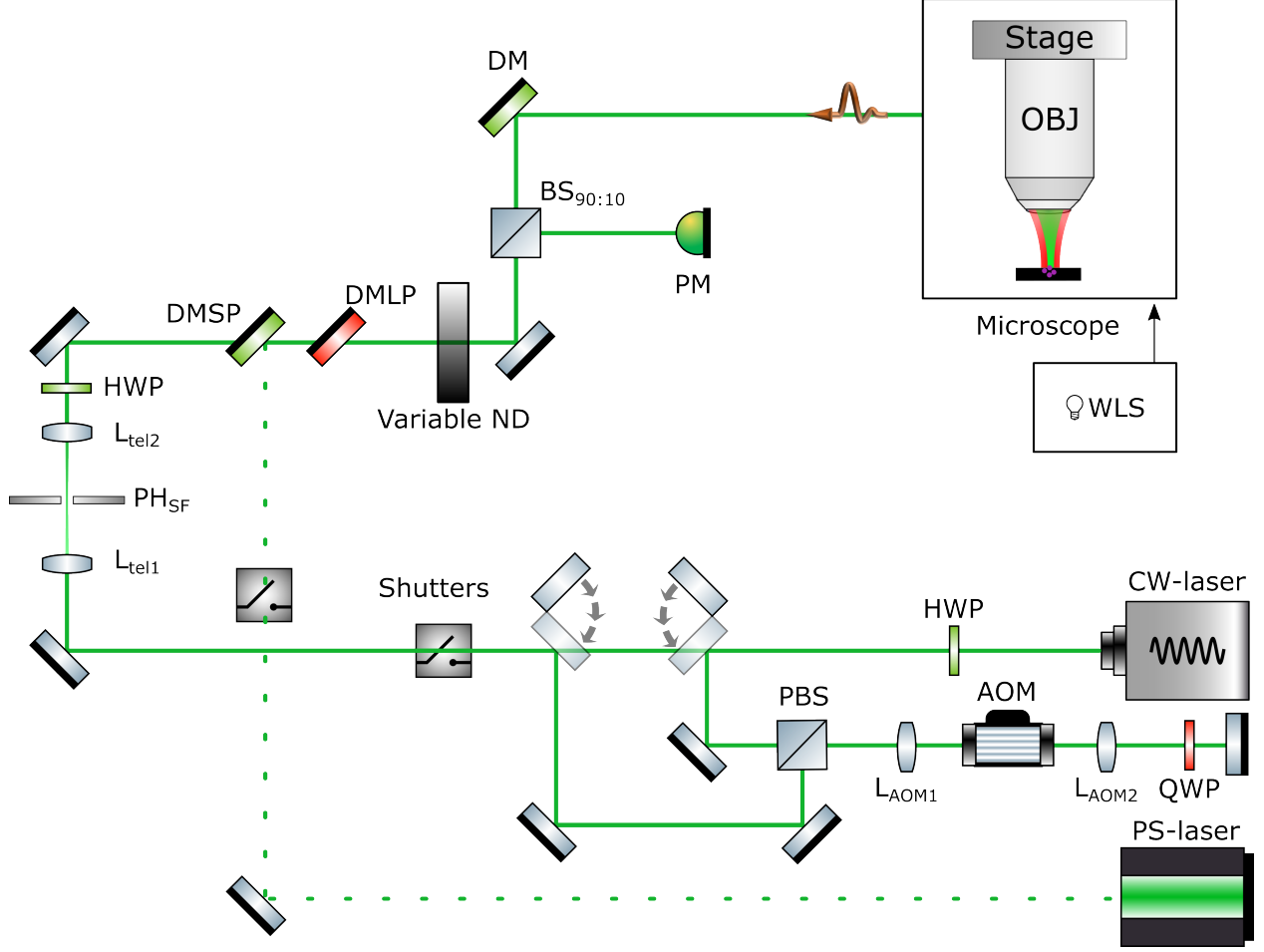


Figure 10: Excitation optical path.

5.2.1 Acousto-Optic Modulator (AOM)

The continuous-wave (CW) 532 nm laser first passes through an AOM, which can be bypassed via flip mirrors when not required. In principle, any laser can be used, but the CW 532 is currently setup for use with the AOM. We use the AOM as a high-speed shutter with a ~ 9 ns rise time to generate the nanosecond pulse sequences necessary for coherent spin manipulation, specifically for optically detected magnetic resonance (ODMR), T_1 , T_2 , and Rabi oscillation measurements.

The AOM is aligned in a double-pass configuration. This geometry explicitly eliminates "beam walk", which is the frequency-dependent spatial shift of the first-order diffracted beam. By retro-reflecting the diffracted beam back through the crystal, the angular deflection is compensated for and we get increased diffraction efficiency. This setup utilizes a quarter-wave plate ($\lambda/4$ or QWP) and a polarizing beam splitter (PBS). The beam passes through the QWP twice (before and after the retro-reflection mirror), rotating its linear polarization by exactly 90° . This allows the PBS to efficiently separate the modulated output beam from the input path.

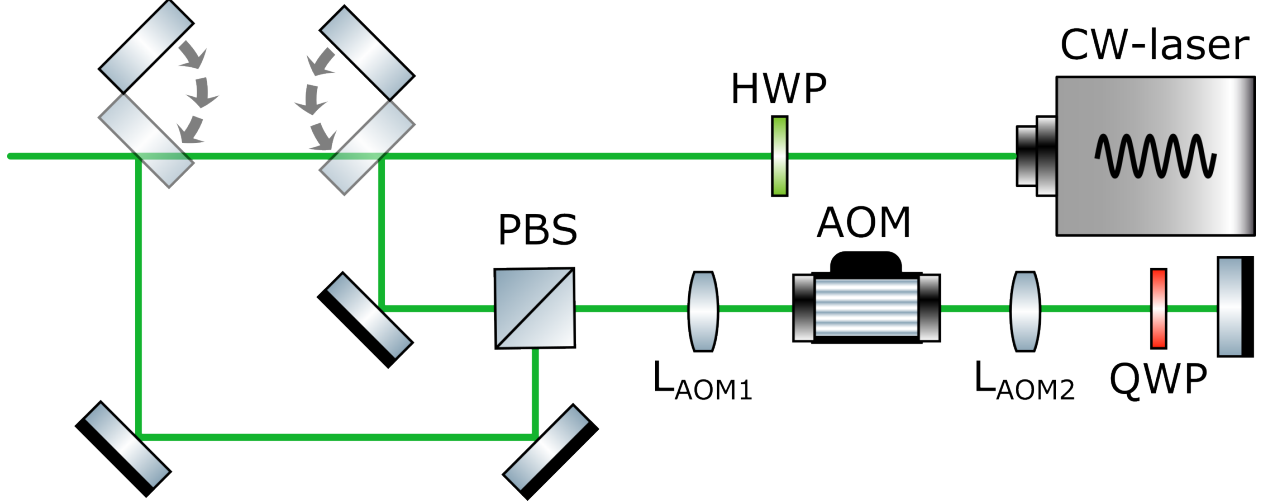


Figure 11: Double-pass AOM optical path.

A double-pass AOM is challenging to align, which is why flip-mirrors are installed to bypass the path entirely when modulation is not required. This configuration was initially aligned following the procedure described by McCarron [2]:

1. Ensure the incident laser beam is completely horizontal and aligned parallel to the optical table's screw-hole grid. This simplifies the longitudinal translation of lens L_{AOM2} in later steps.
2. Position L_{AOM1} such that the beam passes precisely through its optical center. This is verified if the downstream beam position remains strictly unchanged when the lens is inserted or removed.
3. Locate the beam waist via knife-edge measurements. The $1/e^2$ beam radius (w_x) is determined by measuring the translation distance (X_{10-90}) required for the beam power to drop from 90 % to 10 % using the relation:

$$w_x = \frac{X_{10-90}}{1.28} \quad (2)$$

The required focal length (F) for L_{AOM1} to achieve the target waist diameter (d) specified by the manufacturer is estimated by $d = 1.27F\lambda/D$, where D is the $1/e^2$ diameter of the input beam. For this setup at $\lambda = 532$ nm, achieving the target beam diameter of $50 \mu\text{m}$ (specified in Table 11) with an input beam diameter of 1.35 mm results in a required focal length of approximately 100 mm.

4. Turn on the AOM RF driver and position the crystal at the beam focus, accounting for the optical path length of the TeO_2 crystal (usually on the order of millimeters). Adjust the Bragg angle of the AOM to achieve maximum first-order diffraction efficiency.
5. Turn off the AOM. Insert L_{AOM2} one focal length away from the AOM center, ensuring the diverging beam is centered on the lens to recollimate it.

6. Insert a $\lambda/4$ waveplate and a retro-reflecting mirror at normal incidence after L_{AOM2} , centering the waveplate on the transmitted zeroth-order beam.
7. Place an alignment aperture at the input to L_{AOM1} to act as a spatial guide for the return beam.
8. Retro-reflect the zeroth-order beam using the mirror, ensuring it passes back precisely through the input aperture.
9. Place a power meter at the output of the PBS. Maximize the rejected power by fine-tuning the retro-mirror tilt and iteratively rotating the $\lambda/4$ waveplate.
10. Turn the AOM back on and translate the $\lambda/4$ waveplate laterally to intercept and transmit the first-order diffracted beam. Position an aperture between L_{AOM2} and the waveplate to act as a spatial filter, blocking the zeroth-order light and transmitting only the first-order diffracted beam.
11. Make final fine-tune adjustments to the retro-mirror to maximize the overall double-pass efficiency measured at the PBS output.
12. Optimize the longitudinal position of L_{AOM2} to strictly minimize beam deflection as the AOM frequency is swept. Observe the double-passed output beam at a known, long distance downstream. Measure the lateral deflection of the beam (using a knife-edge to find the center) across the RF frequency range of interest (e.g., 330 MHz to 370 MHz). Plot this deflection as a function of the lens position. The optimal position for L_{AOM2} corresponds to the absolute minimum of this deflection curve.

Relevant specifications for the Gooch & Housego 3350-199 AOM are detailed below in Table 11:

Table 11: Gooch & Housego 3350-199 AOM Specifications at 532 nm.

Specification	Value
Operating Wavelength	532 nm
AO Medium	TeO ₂
Acoustic Velocity	4.2 mm/ μ s
Active Aperture	0.15 mm (H) $\times$ 2.0 mm (L)
Center Frequency (F_c)	350 MHz
Target Beam Diameter	50 μ m
Diffraction Efficiency	80 %
Rise Time	9 ns
Bragg Angle	22.2 mrad
Beam Separation	44.4 mrad
Maximum RF Power	1.00 W

5.2.2 4f System: Spatial Filter & Telescope

Before entering the microscope, the laser beam must be spatially cleaned and collimated. We use a $4f$ optical system (Fig. 12) that serves both as a spatial filter and a beam expander. A lens naturally performs a two-dimensional spatial Fourier transform of an incident field at its focal plane. By placing two lenses (L_1 and L_2) a distance $f_1 + f_2$ apart, the system forms a $4f$ relay with access to the Fourier plane at the common focal point.

At this plane, the beam's propagation angles are mapped to transverse physical locations. The low spatial frequencies exist near the optical axis, while high spatial frequencies exist further from the center. As illustrated in Fig. 13A, the input beam may contain high-spatial-frequency artifacts such as interference fringes, scattering, or higher-order transverse modes. After the first lens, these components are spatially separated in the Fourier plane (Fig. 13B). The fundamental Gaussian mode (TEM_{00}) forms a bright central spot on-axis, while higher-frequency distortions are displaced outward. A $25\text{ }\mu\text{m}$ pinhole placed at this Fourier plane acts as a low-pass spatial filter (Fig. 13E), transmitting only the central Gaussian component and physically blocking the high-frequency noise.

The second lens performs the inverse Fourier transform, reconstructing the beam in the output plane. As shown in Fig. 13F, the initially distorted beam profile is converted into a clean, near-Gaussian mode. The trade-off is reduced optical power due to the rejected spatial components (transmitted energy decreases from 1.00 to 0.83 in the simulation). Because the pinhole truncates the field with a hard circular aperture, weak Airy diffraction rings are introduced (note: they are barely visible in this simulation). In practice, these rings are clipped by an iris or the objective back aperture and do not negatively affect the final focus at the sample.

In addition to spatial filtering, this $4f$ configuration functions as a Keplerian telescope. The magnification is set by the focal length ratio:

$$M = \frac{f_2}{f_1}$$

With $f_1 = 75\text{ mm}$ and $f_2 = 100\text{ mm}$, the system expands the beam by $M = 1.33$. This controlled expansion allows the beam diameter to properly match the objective's back aperture, ensuring efficient filling of the numerical aperture (NA) and optimal focusing performance. The second lens (L_2) is mounted on a translation stage along the optical axis, enabling fine adjustment of the output collimation before the beam enters the microscope.

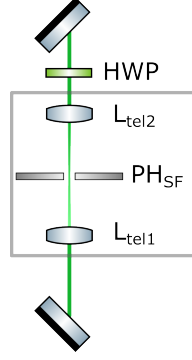


Figure 12: Schematic of the $4f$ spatial filter and beam expander.

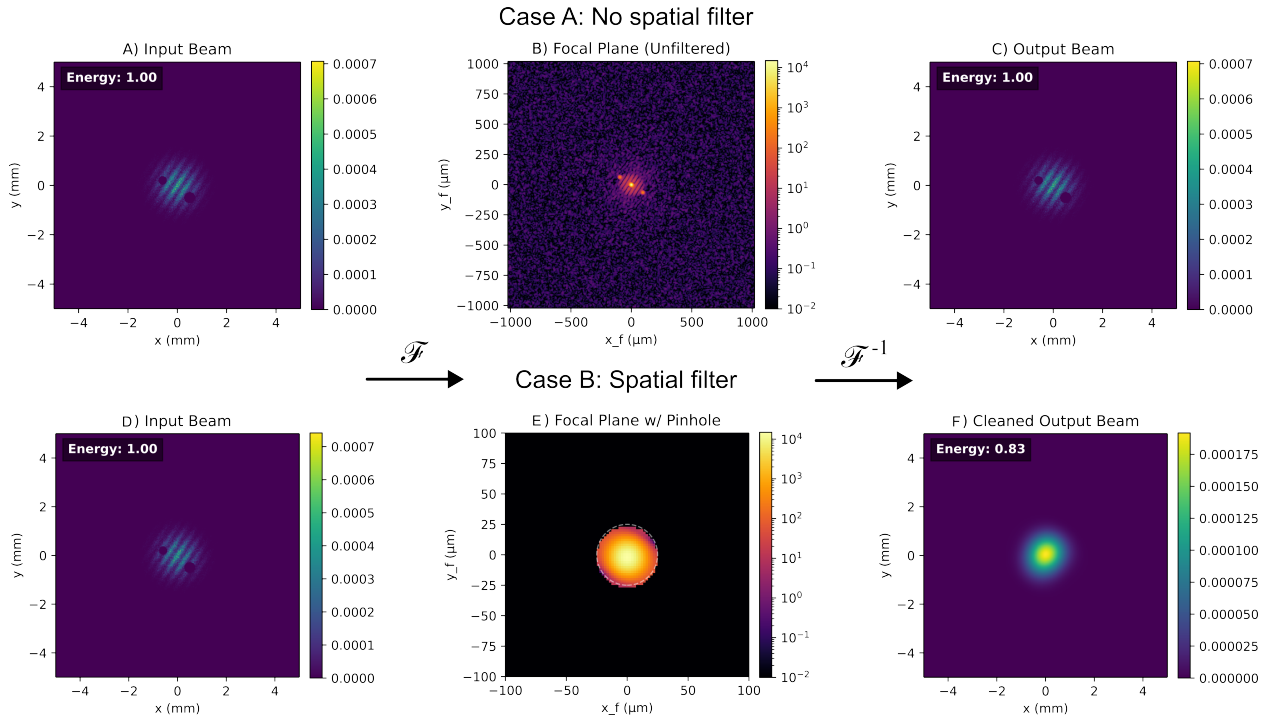


Figure 13: Simulation of low-pass spatial filtering. **Top row:** A Gaussian beam is intentionally degraded with dust-like obscurations, interference fringes, and high-frequency speckle. Its Fourier spectrum (center) is inversely transformed to reconstruct the distorted beam (right), conserving total energy. This transformation is equivalent to the beam propagating through an unfiltered $4f$ system. **Bottom row:** A circular pinhole is placed at the Fourier plane to block high spatial frequencies. The filtered output recovers a cleaner, Gaussian-like profile, accompanied by a 17% reduction in transmitted energy.

To align the $4f$ system:

1. **Establish the Optical Axis:** Remove L_1 , L_2 , and the pinhole from the beam path. Use the upstream steering mirrors to walk the beam through two irises placed downstream of the $4f$ location. Ensure the beam is parallel to the optical table (referring to the screw holes) to

eliminate the need for angular corrections later. Do not touch the upstream steering mirrors after completing this step.

2. **Align the First Lens (L_1):** Insert the focusing lens (L_1). Adjust its x/y position until the beam passes through its center. This is verified when the focused, converging/diverging beam remains perfectly centered on your downstream irises.
3. **Locate the Pinhole at the Focal Plane:** Place the 25 μm pinhole on an XYZ translation stage approximately 75 mm (f_1) downstream of L_1 . Place a power meter immediately after the pinhole. Translate the pinhole longitudinally (z) to find the focal plane, and laterally (x/y) to center it on the beam waist, maximizing the transmitted power. Do not adjust L_1 or the upstream mirrors to the pinhole. The pinhole must be moved to the established beam axis.
4. **Insert and Align the Second Lens (L_2):** Place the collimating lens (L_2) approximately 100 mm (f_2) downstream of the pinhole. Adjust its x/y position so that the emerging beam once again passes cleanly through the center of your downstream irises.
5. **Optimize Collimation:** Translate L_2 along the optical axis (z) using the translation stage it is mounted on until the output beam is collimated. Verify it is collimated by using a mirror to send the beam a reasonable distance across the room and seeing that it maintains its diameter.

5.3 Entering the Microscope: Main Dichroic Alignment

The main dichroic mirror directs the excitation beam into the microscope's Right Side Port (RSP). The beam must enter perfectly centered and parallel to the microscope's internal optical axis. This is difficult to achieve because the microscope does not accommodate standard optical irises.

Before attempting this alignment, it is good to understand the difference between displacement and angular errors at the objective's back aperture:

- **Lateral Displacement Error:** The beam is perfectly parallel to the optical axis but enters the objective off-center. The focused spot will still form at the correct X, Y coordinate on the sample, but asymmetric filling of the back aperture will cause vignetting, reduced excitation efficiency, and a slightly degraded focal spot shape.
- **Angular Error:** The beam enters the objective at a tilt. A microscope objective maps input angle to lateral focal position ($\Delta x \approx f_{obj}\theta$). For our 100 $\times$ objective ($f_{obj} = 2\text{ mm}$), an angular error of just 1 mrad ($\sim 0.05^\circ$) shifts the focal spot laterally by 2 μm at the sample plane.

When co-aligning multiple lasers, angular error is a problem. If two lasers have even a slight angular mismatch, they will focus on entirely different physical locations on the sample. Because quantum emitters are sub-micron in size, switching between slightly misaligned lasers will cause

the excitation spot to miss the target. While you should strive for zero angular error, you should perform a quick re-scan to verify emitter location when switching between laser sources.

5.3.1 The “Quasi-Iris” Alignment Method

Because we cannot place standard irises inside the Ti-U microscope, we use two physical references to establish the true optical axis:

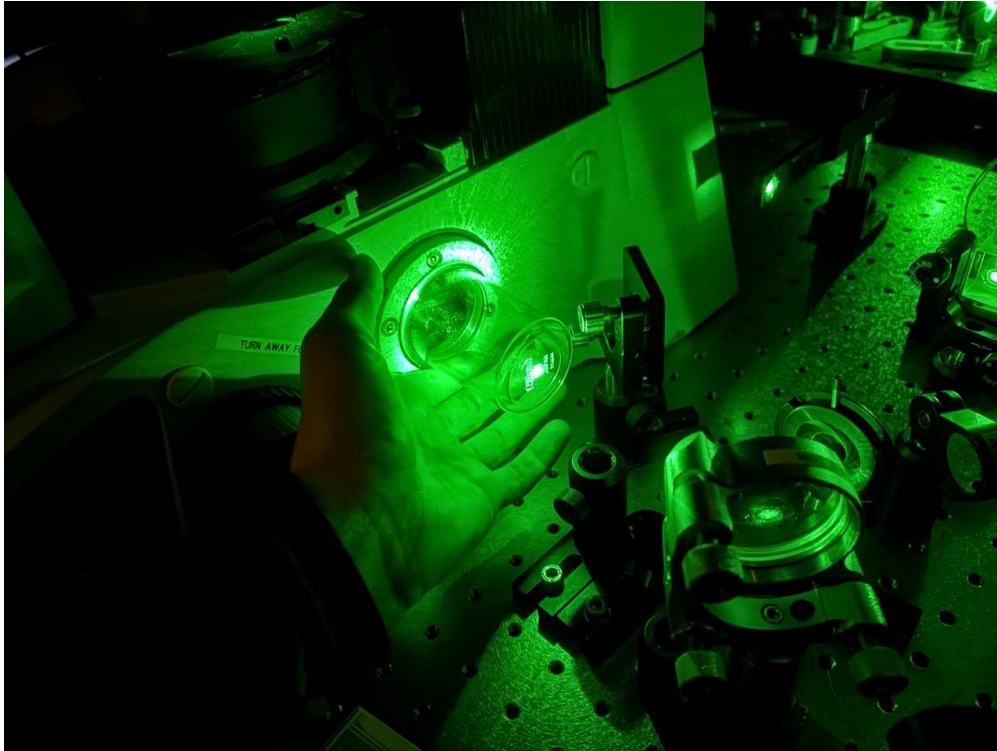


Figure 14: Photograph of the plastic Nikon objective storage container. Its outer diameter matches both the RSP and the objective turret threads, and its molded center dot makes it an ideal “quasi-iris” for establishing the microscope’s internal optical axis.

1. **The Nikon Objective Case (Fig. 14):** The plastic capsule used to store Nikon objectives happens to have the exact same thread pitch and outer diameter as both the RSP entrance and the objective mounting turret. Conveniently, it also features a molded dot at its center. This container acts as our temporary proximal and distal iris.
2. **The Ceiling Mark:** A sticker marked with an “X” is located on the ceiling directly above the microscope. When the objective is removed, this provides a long lever-arm target to guarantee the beam is pointing straight up along the true vertical axis of the system.

Procedure for Aligning the Primary Beam:

1. **Establish Dichroic Position:** Ensure the main dichroic mirror is positioned over the permanent marker outlines drawn on the optical table. These markings are accurate to the

current iris system, but if that system were ever removed, this would only serve as a starting point for the dichroic mirror’s actual optimal position.

2. **Proximal Centering (RSP):** Place the plastic objective container into the opening of the RSP. Adjust the translation of the dichroic mirror mount until the laser hits the exact center dot of the plastic container.
3. **Distal Centering (Turret):** Remove the actual glass objective from the microscope (make sure the stage is off) and put the plastic objective container into the empty turret slot. Adjust the tilt (pitch/yaw) of the dichroic mirror until the laser hits the center dot of the container at the top of the microscope.
4. **Iterate:** Because adjusting the tilt in Step 3 likely shifted the beam position at the RSP, repeat Steps 2 and 3 iteratively until the beam passes cleanly through the center of both “quasi-irises.”
5. **Angular Verification:** Remove the plastic container from the turret. Send the laser straight up through the empty microscope. The beam should hit the “X” sticker on the ceiling.
6. **Establish External Reference Irises:** Once this primary beam is perfectly aligned through the microscope and hitting the ceiling target, set up two standard irises on the optical table *upstream* of the dichroic. Align these irises directly to the laser beam (do not move the laser). These irises now define your permanent, correct optical axis.
7. **Fine alignment of the objective:** Place the objective back into the microscope. Ensure the laser has a clean, gaussian profile coming out of the objective, otherwise, revisit the previous steps.
8. **Place a mirror in the sample plane:** Ensure the mirror is in focus by using the microscope image. Tweak the back reflection of the laser using the main dichroic mirror to go back through the excitation path iris system. The adjustment, if needed, should be small or something is wrong.

Co-aligning Additional Lasers: Once the primary beam has established the correct axis through the external irises, *do not touch the main dichroic*. Use the upstream steering mirrors for the subsequent lasers to walk their beams through those same external irises.

5.4 Confocal Collection and Objective Scanning

Our system utilizes an objective-scanning configuration. The sample remains stationary while a Physik Instrumente (PI) P-561 piezo stage translates the objective in X , Y , and Z (travel range: $100\text{ }\mu\text{m} \times 100\text{ }\mu\text{m} \times 100\text{ }\mu\text{m}$) for high-speed analog scanning as discussed in Section 2.3.

This scanning configuration relies on the shift-invariance of an infinity-corrected microscope. Fluorescence exiting the objective is highly collimated. When the objective translates laterally

during a scan, this collimated emission beam is displaced laterally within the infinity space, but its propagation angle remains perfectly parallel to the optical axis. Because a lens focuses all parallel rays to the same point in its focal plane regardless of their lateral offset, the stationary tube lens continues to focus the emission perfectly into the stationary confocal pinhole, as was shown in Section 3.

In contrast, galvanometer-based systems scan by angularly deflecting the excitation beam into a fixed objective. The resulting fluorescence emission returns through the objective at scan-dependent angles. To prevent this angled emission from physically missing the stationary pinhole, the returning light must be optically “descanned” by reflecting off the same moving galvanometer mirrors before reaching the detector. Objective scanning avoids this complexity entirely because no angular deflection is ever introduced into the beam path.

A key advantage of the infinity-corrected objective is the partial decoupling of the excitation and collection alignments. Although both paths share the same high-NA objective, their geometric constraints differ fundamentally:

- **The Excitation Path** is defined by the collimation and angle of the laser at the objective back aperture. As discussed in the previous section, angular errors here shift the focal spot in the sample plane.
- **The Collection Path** is defined entirely by the optical axis of the tube lens and the physical X, Y, Z position of the confocal pinhole.

Fluorescence is emitted by the sample over a broad angular distribution, and the objective collects the portion that falls within its numerical aperture (NA). This collected light is relayed by the tube lens to the confocal pinhole, which strictly defines the detection axis. Because the detection axis is defined solely by the tube lens and pinhole, the collection path can be aligned independently of the excitation path.

5.5 The Light-Tight Detector Box

The confocal pinhole serves as the only optical entrance to the light-tight detector box. Inside this enclosure, the light transmitted through the pinhole is recollimated, routed via beam splitters and flip mirrors, and finally focused by individual lenses onto the active areas of three single-photon detectors and a spectrometer.

All coupling in this box is free-space; there are no fiber optics involved. When aligning these detectors, it is important to understand their physical tolerances. The Excelitas “SPCM” has a relatively forgiving active area of $180\text{ }\mu\text{m}$ and is the easiest to align. Because this detector is also used for primary raster scanning, its optical path should be aligned first. The two MPD “PDMs” feature a much smaller $50\text{ }\mu\text{m}$ active area, making them more difficult to align. The Ocean Optics spectrometer is aligned last.

The general principle for efficient free-space coupling is that the collimated beam directed towards the focusing lenses must be well-centered and propagate exactly parallel to the optical axis

of the lens (i.e., normal incidence). If you find yourself having to translate a focusing lens a large lateral distance to hit a detector's active area, you are compensating for an angular error. It is always better to re-center the upstream mirror or beam splitter so that extreme lens offsets are not necessary.

The following procedure is for a total realignment of the detector box (e.g., if upstream optics were entirely rebuilt). If you only suspect a minor misalignment on a single detector, use common sense: attempt to tweak the XYZ stage of the focusing lens immediately in front of that specific detector before attempting this full procedure.

Note: I typically leave a sample with a mirror, dye, and NV centers on it above the computer monitor in a plastic box labeled "Brandon alignment sample".

5.5.1 Full Detector Box Alignment Procedure

Phase 1: Preparation and Laser Rough Alignment

1. Power off all three SPADs (the SPCM and both PDMs). Physically block the entrance apertures of the detectors with an opaque material. The next steps use visible laser light, which will instantly vaporize the SPADs if they are powered on and unblocked (not really but you should act like that may happen).
2. Remove the individual focusing lenses positioned immediately in front of each detector.
3. Place a highly reflective mirror at the sample plane of the microscope. Allow a attenuated amount of laser light to leak through the main dichroic mirror and pass into the detector box through the confocal pinhole, as long as the detectors are blocked.
4. Adjust the X, Y, Z position of the confocal pinhole until you visually observe maximum laser power transmitting through it.
5. With the pinhole roughly aligned, use the flipper mirrors and beam splitters to route the beam to each blocked detector. Use the optical table's threaded hole grid as a spatial reference to ensure the beam is traveling perfectly parallel to the table. The fixed mechanical height of the detectors is already designed to match the height of the confocal pinhole (within ~ 1 mm). The goal is to ensure the beams are centered on the blocked detector apertures at normal incidence. Small displacement errors will be corrected by the focusing lenses later.

Phase 2: Fluorescent Dye Intermediate Alignment

6. With the rough laser alignment complete, replace the sample mirror with a slide of bright fluorescent dye. Note: I typically leave a sample with a mirror, dye, and NV centers on it above the computer monitor in a plastic box labeled "Brandon alignment sample".
7. **Re-insert all long-pass emission filters** into the optical path to strictly block the laser line.

8. Visually focus the microscope on the dye and switch the output to the RSP. Unblock and power on the detectors.
9. Re-insert the focusing lens in front of the SPCM on its XYZ translation stage. Monitor the SPCM count rate via the LabVIEW interface.
10. Adjust the SPCM focusing lens in X, Y, Z until you observe a signal. Iteratively maximize the count rate by tweaking the lens position, and then gently tweaking the pinhole X, Y position.
11. Repeat the lens insertion and XYZ optimization for both PDMs to maximize their respective count rates from the bright dye source. This provides an excellent intermediate alignment that makes isolating a single emitter much easier.

Phase 3: Point-Source Fine Alignment

12. Replace the dye slide with a known point-source sample (e.g., a substrate with nanodiamond NV centers). Perform a confocal scan using the SPCM. The emitters may appear slightly out of focus, but should be detectable.
13. Select a bright NV center in the software. Adjust the objective piezo Z -axis by a few microns until the count rate is strictly maximized (ensuring the emitter is at the exact focal plane).
14. Return to the detector box. Iteratively tweak the X, Y position of the SPCM focusing lens and the confocal pinhole until the single-emitter count rate reaches its absolute maximum.
15. Finally, fine-tune the X, Y, Z stages of the two PDM focusing lenses until their count rates are maximized and roughly equivalent (accounting for the splitting ratio of the beam splitter feeding them). This step should only require micro-adjustments if Phase 2 was performed correctly.

5.5.2 Spectrometer Alignment

For the Ocean Optics spectrometer, follow the same principles outlined above: start with attenuated visible laser light while the spectrometer entrance is blocked to establish a rough axis.

For fine alignment, open the Ocean Optics control software and start the live viewer with an integration time of ~ 500 ms. Use the fluorescent dye sample with a high laser excitation power. Do not attempt to align the spectrometer using an NV center point-source, as the single-photon count rates are typically too low to easily resolve on the spectrometer in real-time without impractically long integration periods. Maximize the live spectral intensity by translating the spectrometer's focusing lens in X, Y, Z .

6 System Operation

6.1 Powering on devices

1. Turn on the PI piezo stage (large controller on the rack above the table).

2. Turn on the 532 nm laser (in the Ltune software on the desktop). Set the "Modulation Mode" to "Internal Constant".
3. Turn on the SPCM detector (power supply on rack above table, should be set to 5V).
4. Other lasers are optional, but Mai Tai is turned on from the software on the laptop connected to it and Becker & Hickl is turned on by the key directly on its controller.

6.2 Loading Your Sample

Place your sample on a glass slide using double-sided tape, then mount the slide onto the microscope stage. There are three primary microscope body dials to configure, and you only need to memorize two specific configurations for standard operation.

For direct observation through the eyepieces, the microscope should be configured with the front selector set to **Eyepiece**, the side-port selector set to **Eye**, and the filter-wheel selector set to **BS**. Remember to turn the white light source on. For confocal measurements using the excitation laser and external detectors, the microscope should be configured with the front selector set to **Right**, the side-port selector set to **R**, and the filter-wheel selector set to **DF**.

Table 12: Microscope selector settings for visual observation and confocal measurements.

Operating Mode	Front Selector	Port Selector	Filter Wheel
Visual observation through eyepieces	Eyepiece	Eye	BS
Laser excitation and detector measurements	Right	R	DF

Note on the Focusing Mechanism and Tension Control The Nikon Ti-U utilizes a nosepiece-elevation focusing system where rotating the knobs moves the objective up or down. To ensure stable imaging and prevent mechanical wear, you must utilize the adjustment rings located behind the focus knobs:

- **Torque Adjustment (Left Side):** The left coarse focus knob is equipped with a torque adjustment ring. Rotate this ring **counter-clockwise** to tighten the tension; this is necessary to prevent the nosepiece from lowering under its own weight and losing focus. Rotate it **clockwise** to loosen the movement.
- **Coarse Focus Stopper (Right Side):** The right knob features a stopper ring used to "lock" the upper limit of the coarse travel. Once you have found the focal plane of your specimen, rotate this ring **clockwise** to the limit. This allows you to lower the objective to switch samples and return to the approximate focal plane by simply rotating the coarse knob back until it hits the physical stop.

Never attempt to rotate the left and right focus knobs in opposite directions simultaneously, as this is not great for the mechanism. If you feel significant resistance, check the stopper and torque rings rather than forcing the knobs.

6.3 Software Initialization

The custom control software, pictured below, must be launched from its dedicated LabVIEW project file to correctly call its hardware dependencies.

1. Open the LabVIEW project entitled **TCSPC Controller** located on the desktop. (The source repository is hosted on GitHub).
2. Open `main.vi` and allow it time (up to one minute) to load all resources into memory.
3. Press the white arrow **Run** button in the top-left toolbar. Give the software up to 30 s to initialize all hardware stages and motors.
4. Verify that the piezo stage position, power meter, and motorized filter wheel are all displaying active, live reading values on the front panel.
5. Toggle the **Laser Shutter** button on and off once to wake up the controller.
6. **Set your working directory** at the top right of the interface (Fig. 15). Your acquired data will be saved to this folder path (you will be prompted before saving in manual mode).

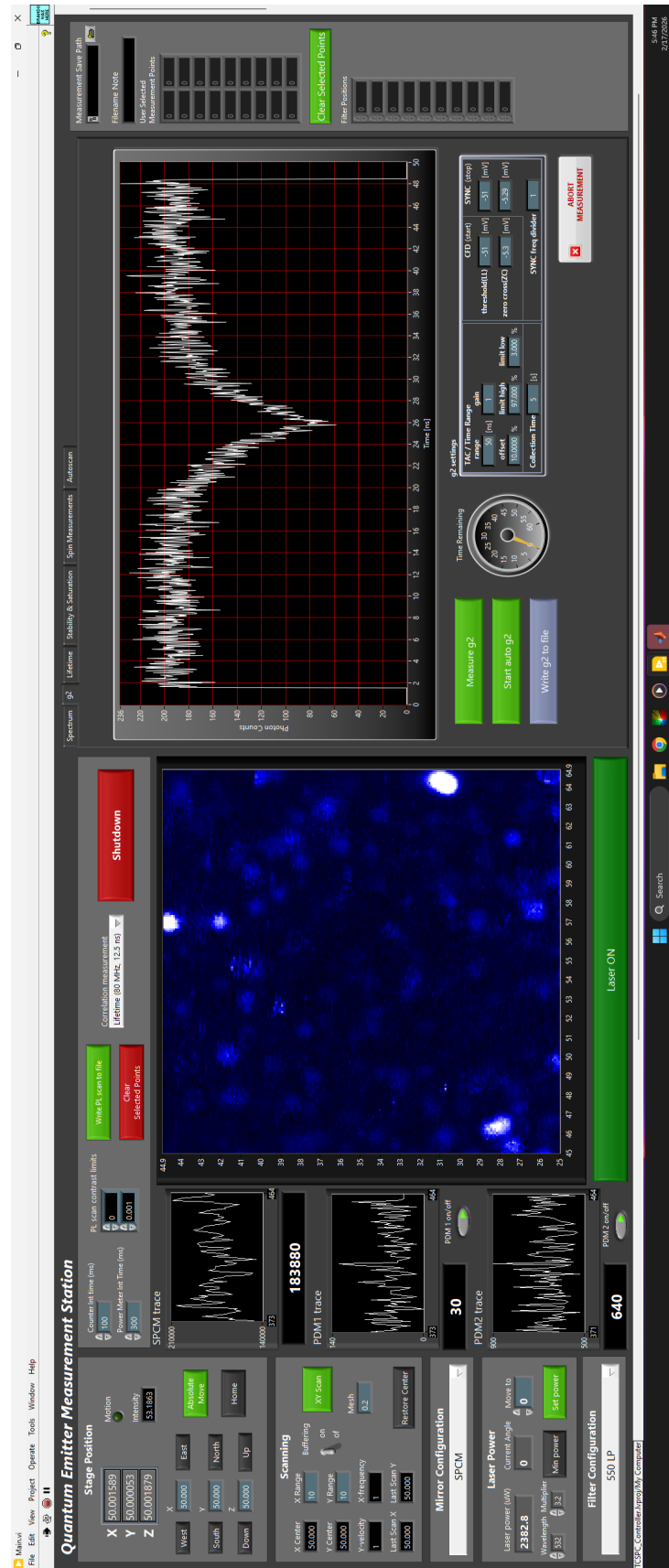


Figure 15: Screenshot of the LabVIEW main control interface.

The standard experimental workflow begins by conducting a rough, large-area PL scan to locate your features of interest, optimizing the optical focus (which is typically $\sim 1\text{ }\mu\text{m}$ axially offset from the visual white-light focus), and then conducting a high-resolution PL scan over the target emitter.

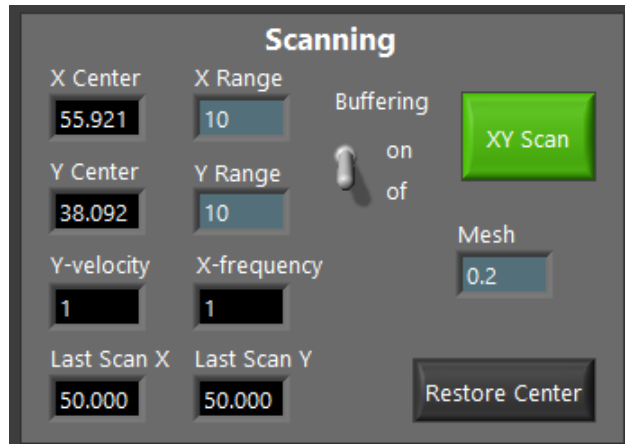
6.4 PL Scan Procedure

All units are in micron. To acquire a photoluminescence (PL) scan of the sample:

1. Press the **Home** button to bring the piezo stage to its center travel position. If your sample moved significantly in Z during sample loading, use the microscope eyepieces to put it back into visual focus, then switch back to confocal mode (Section 6.2).
2. Configure the motorized flip mirrors within the light tight box to direct the beam to the **SPCM** detector, as shown in Fig. 17.
3. Un-shutter the laser with the button in labVIEW.
4. **Set Parameters:** In the **Scanning** section (Fig. 16), set the *X* and *Y* scan range to a reasonable search area (e.g., $30\text{ }\mu\text{m} \times 30\text{ }\mu\text{m}$). The hardware limit is $100\text{ }\mu\text{m} \times 100\text{ }\mu\text{m}$. Set the **Mesh** to a relatively coarse value of $0.3\text{ }\mu\text{m}$ or higher.

Note: The mesh is the step size of the raster scan. It sets the resolution of the resulting image. For a scan of physical size $L \times L$ and mesh size Δx , the number of pixels along each axis is given by $N = L/\Delta x$.

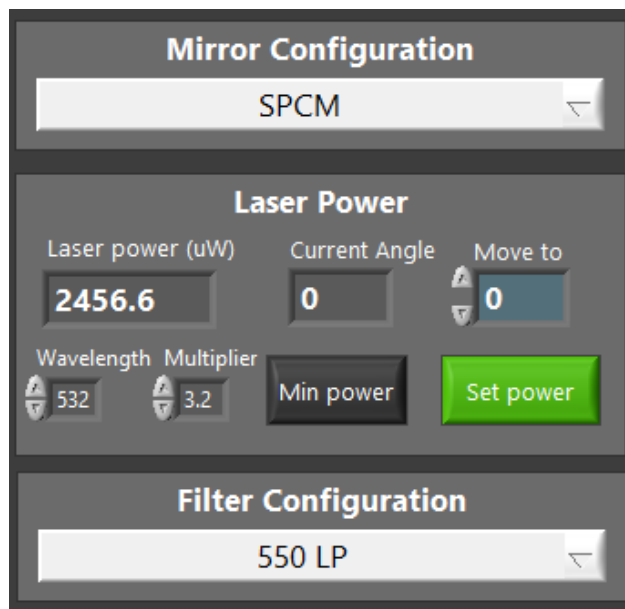
5. **Start Scan:** Press **XY Scan**. At these coarse settings, the acquisition will take approximately 1 min.
6. **Optimize Focus:** When the scan completes, click on a bright, localized feature in the generated plot window and press **Move to Position**. Adjust the objective Z-axis position up and down in small increments until the live photon count rate is strictly maximized.
7. **High-Resolution Scan:** With the optimal focal plane established, repeat the scan using a tighter bounding box and a finer mesh size to acquire highest quality mapping data. Recommended high-resolution parameters are a $10\text{ }\mu\text{m} \times 10\text{ }\mu\text{m}$ area with a $0.05\text{ }\mu\text{m}$ mesh. Meshing finer than this generally oversamples beyond the optical diffraction limit and wastes time.
8. **Save Data:** Press **Write PL scan to file**. The software will prompt you to save, automatically generating a filename that preserves all critical metadata (Laser Power, Filters, Mesh Size, Scan Center Coordinates, etc.).



Scanning

X Center	X Range	Buffering on off	XY Scan
55.921	10		
Y Center	Y Range		
38.092	10		
Y-velocity	X-frequency	Mesh	
1	1	0.2	
Last Scan X	Last Scan Y	Restore Center	
50.000	50.000		

Figure 16: Scanning parameters input area. All units are in micron. The X Center and Y Center are the starting coordinates of your scan. The range is the span of the scan. The Y-velocity and X-frequency should be left as default settings. The Buffering should be left off. Restore Center updates the X Center and Y Center variables to what they were for the last scan taken. Mesh is the step size in micron.



Mirror Configuration

SPCM

Laser Power

Laser power (uW)	Current Angle	Move to
2456.6	0	0
Wavelength	Multiplier	
532	3.2	
		Min power
		Set power

Filter Configuration

550 LP

Figure 17: Mirror routing and filter configuration area.

6.5 Spectrum Measurement

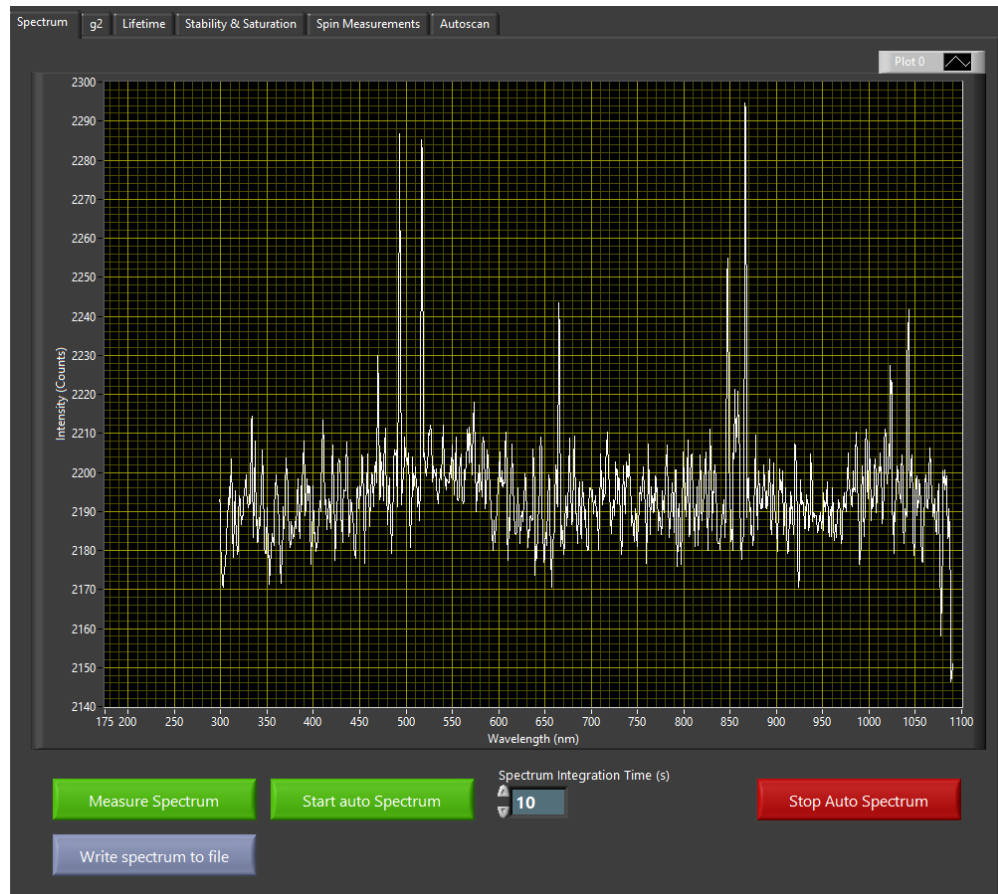


Figure 18: Spectrum measurement tab.

To acquire an optical spectrum after centering the beam on a point of interest:

1. Navigate to the **Spectrum** tab (Fig. 18).
2. Change the mirror configuration (Fig. 17) to route the emission to the **Spectrometer**.
3. Set the desired **Integration Time** based on the expected signal brightness.
4. Press **Measure Spectrum**. *Note: This measurement will briefly disable the LabVIEW block diagram while the Ocean Optics driver acquires the data.*
5. Press **Write spectrum to file** and follow the dialogue prompt to save the data.

6.6 Stability & Saturation Measurements

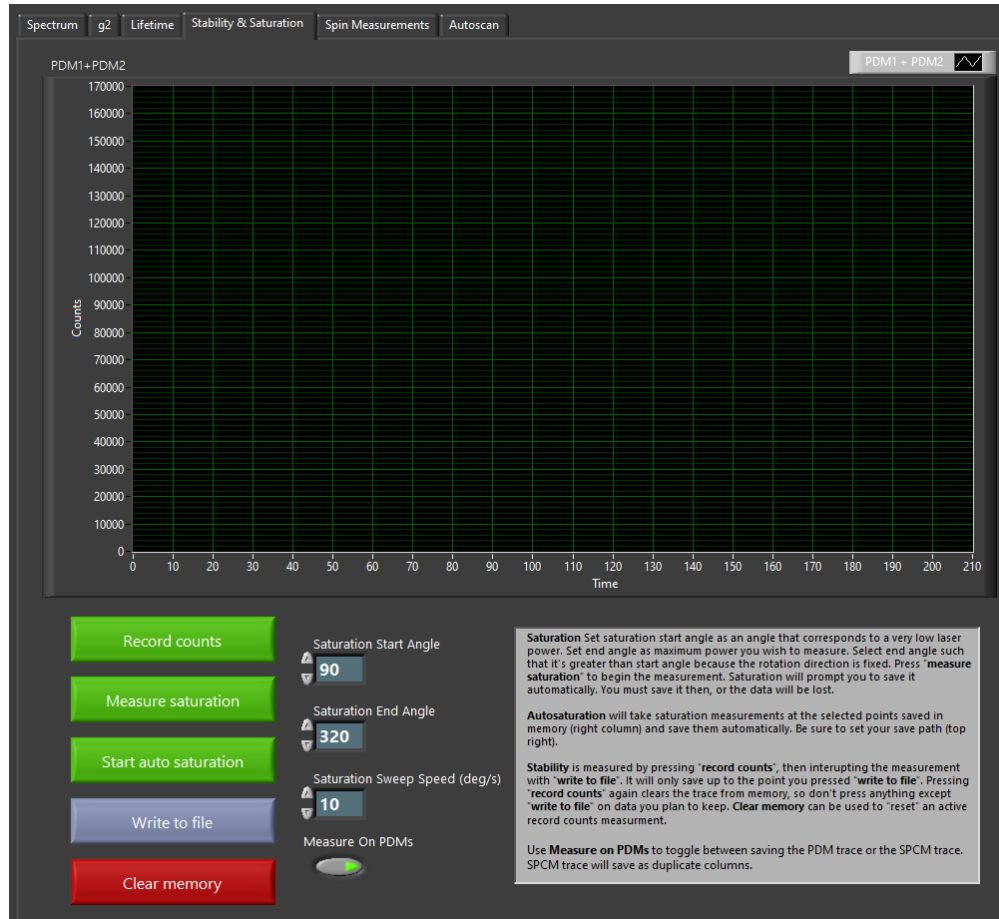


Figure 19: Stability & Saturation measurements tab.

To take a Stability Measurement (after focusing on a point of interest):

1. Navigate to the **Stability & Saturation** tab (Fig. 19).
2. It no longer matters which detector you select to measure on (the note on Fig. 19 is outdated). The data from all three detectors will be summed. This adds a small amount of dark counts to the measurement at the benefit of greatly reducing the chance of error with the measurement.
3. Press **Record Counts** to begin logging the temporal trace.
4. Pressing **Record Counts** again will pause the measurement. To conclude the measurement, press **Write to file** and save the acquired data.

*Note: You can flush the recorded data array while actively taking a measurement (effectively resetting the trace to $t = 0$) by pressing **Clear memory**.*

To take a Saturation Measurement (after focusing on a point of interest):

1. Navigate to the **Stability & Saturation** tab.
2. Change the hardware mirror configuration to match your detector choice.
3. Set your **Saturation Start Angle**, **Saturation End Angle**, and **Saturation Sweep Speed**. The start and end angles correspond to the motorized rotation stage values for your minimum and maximum desired laser excitation powers, respectively. You should test beforehand to see what values correspond to the power range you wish to scan. Ensure the end angle is numerically higher than the start angle.
4. Press **Measure saturation**. The system will automatically sweep the power and log the corresponding count rates.
5. When the sweep is complete, the software will prompt you to save.

6.7 Lifetime Measurement

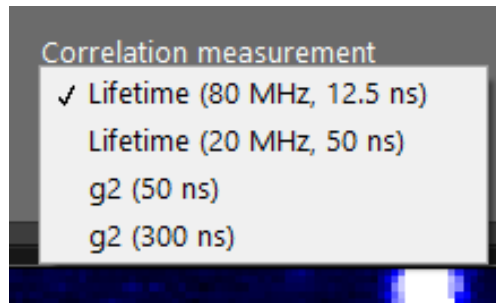


Figure 20: Correlation measurement configuration drop-down menu.

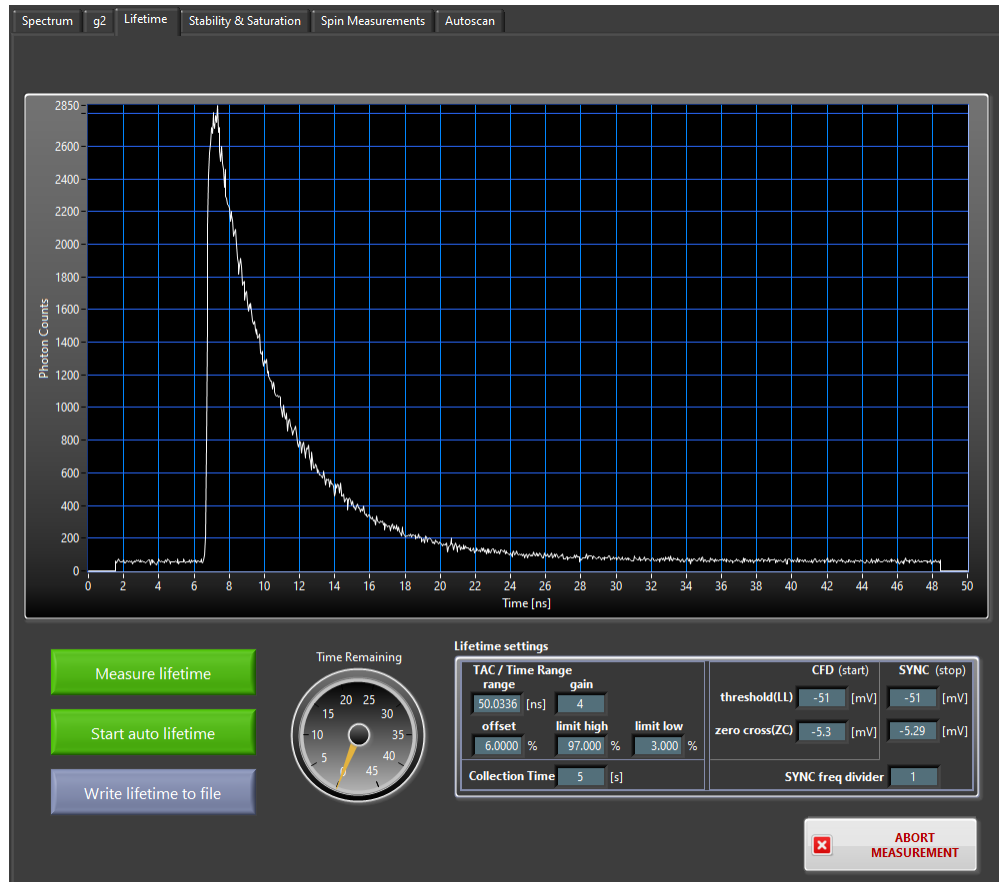


Figure 21: Lifetime measurement tab.

To acquire a fluorescence lifetime measurement, you must physically switch to a pulsed laser source. Turn on the pulsed laser of your choosing (by default, the B&H 515 nm diode should be used). The anticipated lifetime of your specific emitter dictates the required laser repetition rate (e.g., 80 MHz for fast emitters, 20 MHz for slow emitters). Detailed hardware routing for these configurations can be found in Section 4.3.1. These repetition rates are chosen because they are supported by the B&H laser. 2 MHz may be needed for emitters with very long lifetimes, please read Section 4.3.1 if doing this. Because there may be a slight spatial misalignment between the CW laser and the pulsed laser paths, it is highly recommended to perform a quick PL scan using the pulsed laser to re-center the emitter before proceeding.

1. Navigate to the **Lifetime** tab (Fig. 21).
2. Click the **Correlation Measurement** drop-down menu (Fig. 20) and select the hardware timing configuration that matches your physical delay line setup.
3. Selecting this configuration will automatically populate the optimal TCSPC module parameters into the software settings, with the exception of the total collection time. You can overwrite it if needed.

4. Input your desired **Collection Time**.
5. Change the mirror configuration (Fig. 17) to route the emission to the **PDMs**.
6. Press **Measure Lifetime**.
7. The generated histogram will display when the integration completes. Press **Write lifetime to file** to save the raw TCSPC data array.

6.8 $g^{(2)}(\tau)$ Photon Correlation Measurement

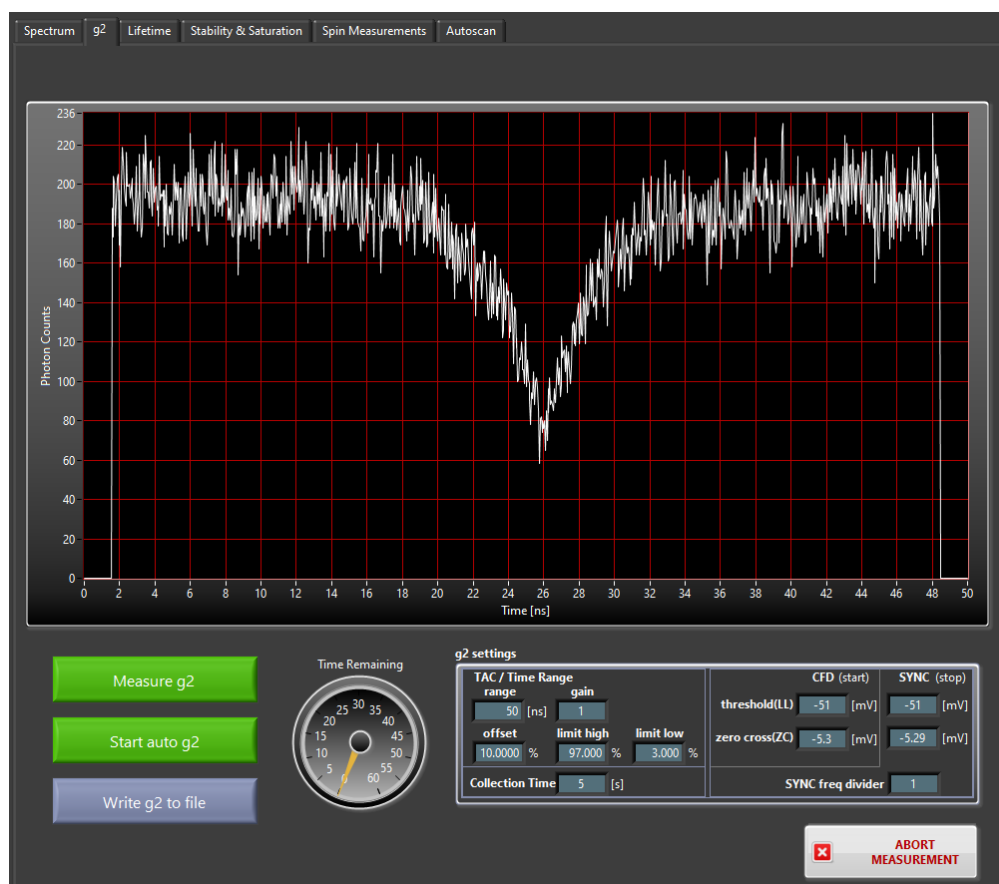


Figure 22: $g^{(2)}$ measurement tab.

To acquire a second-order autocorrelation ($g^{(2)}(\tau)$) measurement, the CW laser is typically used. The specific correlation measurement setting depends on your emitter's photophysics (e.g., shelving states or long lifetimes), but the standard 50 ns TAC window option is appropriate for most cases.

1. Navigate to the **g2** tab (Fig. 22).
2. Click the **Correlation Measurement** drop-down menu (Fig. 20) and select the appropriate $g^{(2)}$ parameter set.

3. Selecting this configuration will automatically populate the optimal TCSPC module parameters into the software settings, with the exception of the total collection time.
4. Input your desired **Collection Time**.
5. Change the mirror configuration (Fig. 17) to route the emission to the **PDMs**.
6. Press **Measure g2**.
7. The generated correlation histogram will display when the integration completes. Press **Write g2 to file** to save the raw TCSPC data array.

6.9 Automated Power Sweeping by Location (Batch Acquisition)

The automated batch acquisition feature functions similarly across all the measurement types. On each measurement panel, there is an option to execute a “Start auto [measurement].” We will use the “auto-g2” routine for this walk-through. The auto- $g^{(2)}$ feature allows you to define an array of specific spatial coordinates and excitation powers, commanding the program to sequentially measure and save the correlation data at each parameter combination automatically.

To execute a batch acquisition:

1. Perform a standard PL scan that encompasses all the physical locations you wish to measure. Ensure the surface is in focus.
2. Press **Clear Selected Points** in the right-hand column (Fig. 23) to flush any old memory arrays.
3. Click directly on the generated PL scan image to select the exact physical locations you wish to measure. You will see the coordinates automatically populate in the **User Selected Points** array.
4. Define the excitation power(s) for the sequence by inputting a motorized rotation stage angle (corresponding to the ND filter position) into the **Filter Positions** array in the right-hand column. You must define at least one power setting.
5. Verify that your **Working Directory** path is set correctly, as the software will generate and save multiple files automatically.
6. Change the mirror configuration to route the emission to the **PDMs**.
7. Press **Start auto g2** to begin the sequence.

The software will automatically append the relevant metadata (power, location, integration time) to the generated filenames. *Note: The “auto-saturation” routine is designed to sweep power, so it will simply ignore any values placed in the Filter Positions (power) array.*

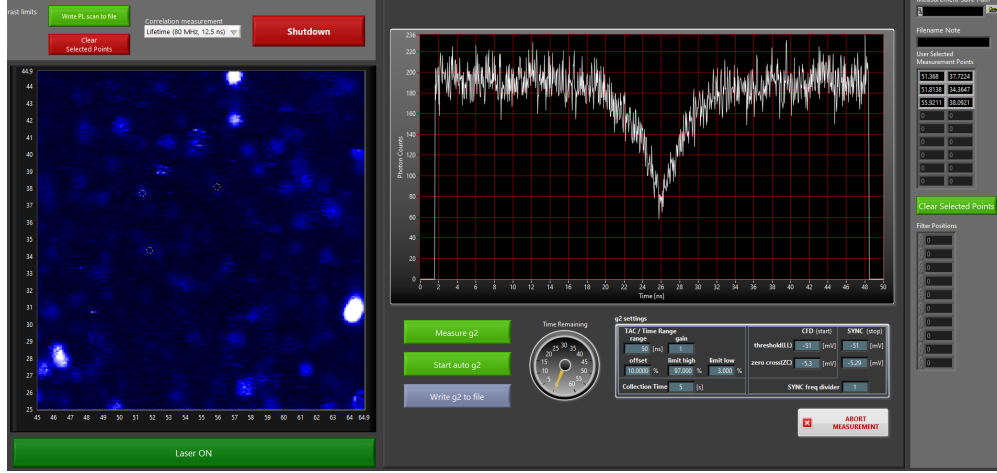


Figure 23: $g^{(2)}$ measurement tab with the batch acquisition column displayed on the right-hand side.

Note regarding Focus Drift: This batch acquisition routine was designed for quick, convenient tests. It is not as robust as the full “auto-scan” feature detailed in the following section. Specifically, it does not utilize the closed-loop auto-focusing algorithm developed by PI to maintain focus over extremely long durations. Therefore, this feature is best suited for tens of measurements at a time (where focus drift isn’t a major issue) rather than multi-day, hundreds-of-points acquisitions.

6.10 Autonomous Emitter Characterization (Autoscan)

The **Autoscan** tab features an algorithm that identifies potential emitters from bright spots on a PL scan, measures them sequentially, and automatically saves the data. This routine includes a closed-loop auto-focusing procedure that executes before each emitter measurement, correcting for mechanical drift to keep the system in focus indefinitely.

The autoscan algorithm operates by dividing a $70\text{ }\mu\text{m} \times 70\text{ }\mu\text{m}$ global area into smaller $10\text{ }\mu\text{m} \times 10\text{ }\mu\text{m}$ sub-regions following a spiral pattern. Alternatively, the user can override this default and manually define specific $10\text{ }\mu\text{m} \times 10\text{ }\mu\text{m}$ search areas. A contrast-based peak detection algorithm analyzes these regions to identify bright spots. Once a candidate spot is identified, the system moves the laser focus to that coordinate, runs a brief search-and-maximize-count routine to optimize laser focus/placement, and then measures the count stability. If the stability falls within the defined RMS error tolerance (the default value of “10000” for “RMS cutoff” in LabVIEW is recommended) and the absolute photon count rate falls within the user-defined bounds, the system validates the emitter and proceeds to conduct all selected measurements (spectrum, $g^{(2)}$, lifetime, stability, saturation).

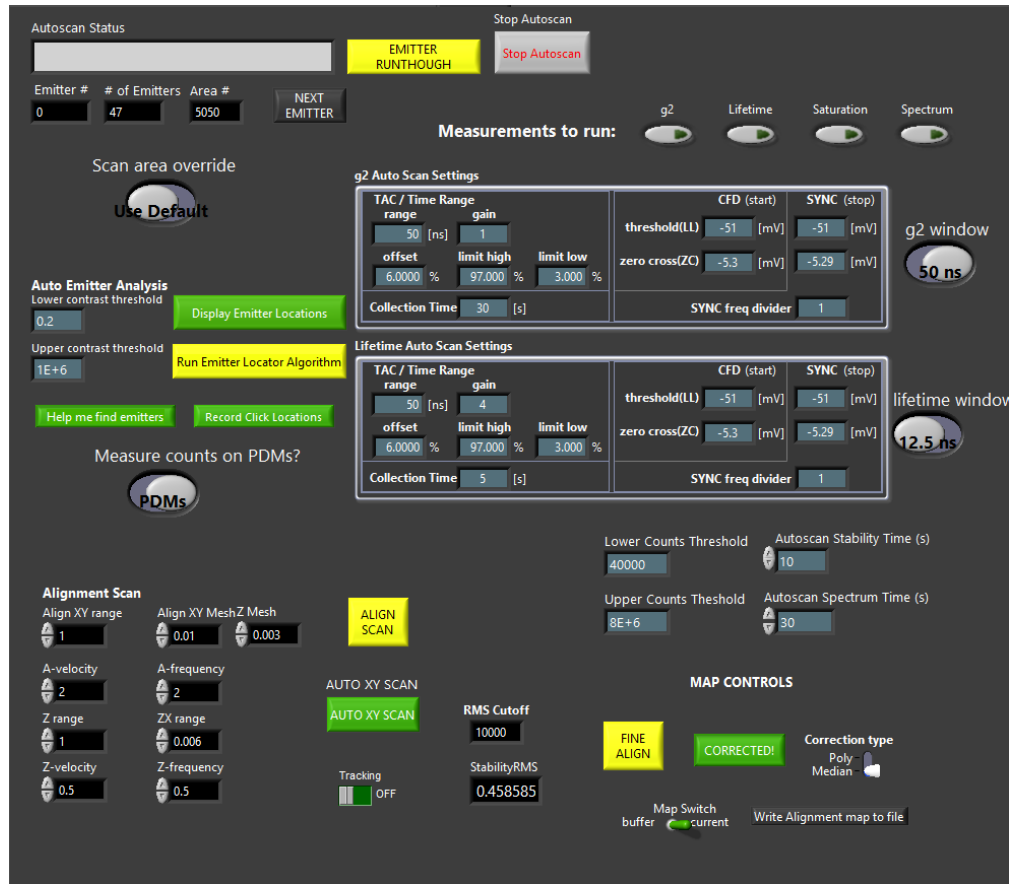


Figure 24: Autoscan measurement tab interface.

6.10.1 Autoscan Configuration Procedure

To configure and launch an automated scanning sequence:

1. **Navigate** to the **Autoscan** tab.
2. **Test Scan:** Take a preliminary PL scan of a $10\mu\text{m} \times 10\mu\text{m}$ area with a fine mesh (no larger than $0.1\mu\text{m}$) to verify emitter presence and set contrast limits for the emitter identification algorithm.
3. **Set Contrast Thresholds:** Input your starting guess for the **Lower contrast threshold** and **Upper contrast threshold**. A good starting point is 0.2 for the lower bound and 1×10^6 for the upper bound.
4. **Run Algorithm:** Press **Run Emitter Locator Algorithm**, then press **Display Emitter Locations**.
5. **Verify Locations:** The UI map will highlight candidate emitter locations.
 - *If too many background points are selected:* Raise the **Lower contrast threshold**.

- *If real emitters are being ignored:* Lower the **Lower contrast threshold**.
6. **Select Scan Area:** Choose either **Default** or **Use user selected** for the **Scan area override** option.
 - *Default:* Autonomously searches and scans the full $70\text{ }\mu\text{m} \times 70\text{ }\mu\text{m}$ travel range.
 - *User Selected:* Restricts the search to specific $10\text{ }\mu\text{m} \times 10\text{ }\mu\text{m}$ bounding boxes centered around the coordinates populated in the **User Selected Points** array (located in the right-hand column).
 7. **Set Count Thresholds:** Define the **Lower Counts Threshold** and **Upper Counts Threshold**. During the preliminary stability test, the emitter's average count rate must fall between these two bounds for the measurements to proceed. Recommended starting values are 40 000 (Lower) and 8×10^6 (Upper).
 8. **Select Measurements:** Check the corresponding boxes in the **Measurements to run** section for every measurement type you want the system to perform on each valid emitter.
 9. **Configure $g^{(2)}(\tau)$:** Set the target $g^{(2)}$ window (Default: 50 ns; toggle to 300 ns if using the long TAC configuration). This will automatically command the NI relay to route the hardware delay lines and populate the optimal TCSPC settings. **Input your desired Collection Time.**
 10. **Configure Lifetime:** Set the Lifetime window (Default: 50 ns; toggle to 12.5 ns if using the short TAC configuration). Ensure your physical pulsed laser is turned on and properly aligned. **Input your desired Collection Time.**
 11. **Configure Other Timers:** Input your desired integration limits into **Autoscan Spectrum Time (s)** and **Autoscan Stability Time (s)**.
 12. **Launch:** Open the physical laser shutter and press **Emitter Runthrough**.

The system status will update in the **Autoscan Status** indicator. You can click through the individual measurement tabs to view the data accumulating in real-time. *Note that the LabVIEW block diagram will temporarily freeze during spectrometer measurements as it waits for the Ocean Optics hardware response.*

6.10.2 Exiting Autoscan

The exit procedure for the Autoscan is not instantaneous. Do not force-quit LabVIEW, as this can crash the hardware drivers. To stop the routine cleanly:

1. Uncheck all boxes in the **Measurements to run** section.
2. Press the **Stop Autoscan** button.

3. **Wait.** The system must finish its current hardware execution loop before it can safely shut down. It will exit within 1 min to 2 min and update the **Autoscan Status** indicator when it is safe to proceed.

7 Data Processing and Analysis

The system includes a Python-based analysis suite for processing the collected data. This software automatically reads, organizes, and plots the various CSV files generated during an experiment, including PL maps, spectra, $g^{(2)}$ measurements, fluorescence lifetime data, and saturation curves. To make this possible, the LabVIEW acquisition software saves important experimental settings directly in each filename. The Python scripts then read this information and use it to process the data without requiring manual input. The source code is hosted on GitHub.

7.1 File Naming Convention and Metadata Extraction

To allow the analysis software to operate automatically, all data files are saved using a consistent filename format. Each filename contains a series of short labels and values that describe how the measurement was performed, such as the sample position, laser power, integration time, and detector settings. These values are recorded by the LabVIEW software at the time of acquisition and embedded directly into the filename.

The general filename format is:

$$[\text{Type}]_ [\text{Param}]_1_ [\text{Param}]_2_ \dots_ [\text{Param}]_n. \text{csv}$$

Example 1 (Correlation): `g2_X50.0_Y52.0_Pow110_Int60_TAC50_Gain1_Bin1024.csv`

This filename indicates a $g^{(2)}$ measurement taken at the stage coordinates (50.0, 52.0) using 110 μW of excitation power and an integration time of 60 s. The TAC range (T_{TAC}), gain (G), and number of bins (N_{bins}) are used to determine the time represented by each histogram bin (Δt_{bin}):

$$\Delta t_{\text{bin}} = \frac{T_{\text{TAC}}}{G \cdot N_{\text{bins}}} \quad (3)$$

Example 2 (PL Mapping): `PL_X54.5_Y55.0_ND0_Pow100_Filter550LP_Size10_Mesh0.05.csv`

This filename identifies a photoluminescence (PL) scan centered at coordinates (54.5, 55.0) with a $10\ \mu\text{m} \times 10\ \mu\text{m}$ field of view and a scan step size of $0.05\ \mu\text{m}$, acquired at 100 μW using a 550 nm long-pass emission filter.

Table 13 summarizes the standard labels recognized by the analysis software.

Table 13: Filename labels used for automated metadata extraction.

Label	Meaning	Units/Format	Used In
Type	Measurement type	PL, Life, g2, Spec	All
X / Y	Stage coordinates	μm	All
ND	Neutral density angle	Degrees ($^\circ$)	All
Pow	Excitation power	μW	All
Filter	Emission filter	Wavelength + type	All
Int	Integration time	s	Spec, Life, $g^{(2)}$
TAC	TCSPC time window	ns	Life, $g^{(2)}$
Gain	TAC gain setting	Unitless	Life, $g^{(2)}$
Bin	Number of histogram bins	Integer	Life, $g^{(2)}$
Size	Scan field of view	μm	PL Scan
Mesh	Scan step size	μm	PL Scan

7.2 Automated Metadata Parsing and Measurement Association

The analysis software begins by reading the filename and extracting the experimental settings listed in Table 13. These settings are then stored in a structured format that can be used throughout the analysis.

The sample coordinates (X , Y) serve as a unique identifier for each emitter. By comparing these coordinates, the software automatically groups together all measurements collected from the same location, including PL maps, spectra, lifetime traces, $g^{(2)}$ correlations, and saturation curves. This creates a single organized record for each emitter and eliminates the need to manually match files across different experiments.

7.3 Interactive Data Exploration (app.py)

For real-time data review and quality control, the suite includes an interactive dashboard.

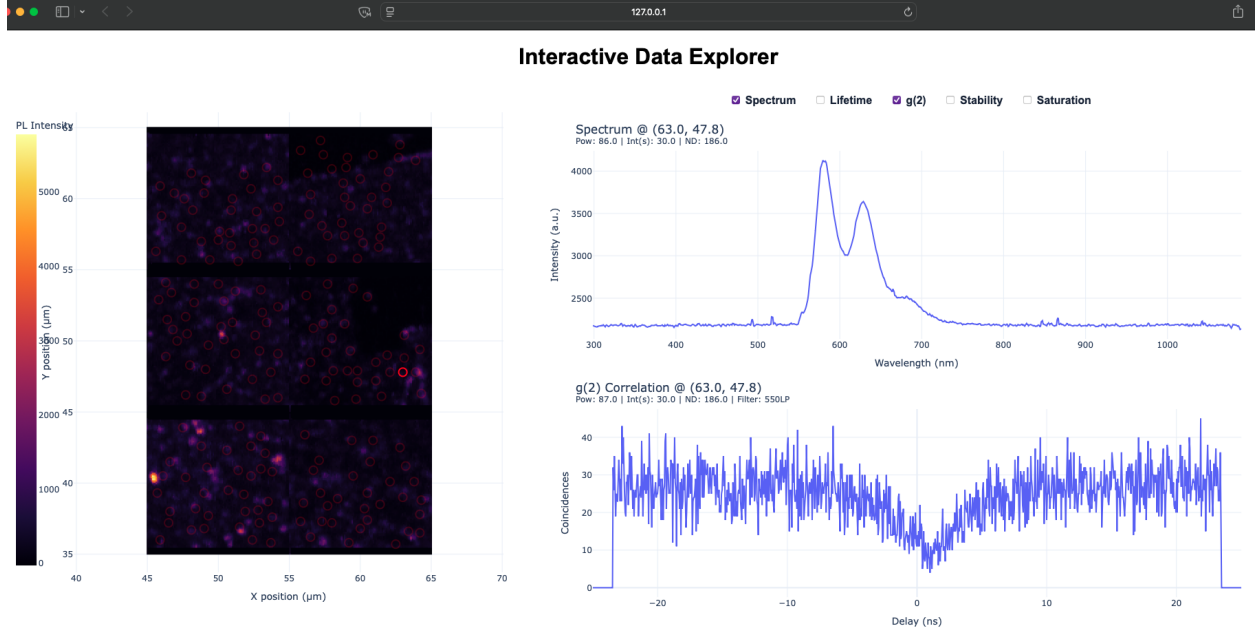


Figure 25: The TCSPC Data Explorer dashboard. Users can click on any coordinate in the stitched PL map to instantly view the corresponding spectrum, $g^{(2)}(\tau)$, and lifetime decay curves measured at that location. Data shown is from a recent carbon-12 doped hBN emitter.

The dashboard provides an interactive view of the experiment:

- **Stitched PL Map:** A global map comprising all individual 10×10 and 70×70 μm scans found in the directory.
- **Point-and-Click Correlation:** Clicking any pixel on the global map queries the database and updates a 2×3 grid of subplots displaying every measurement (Spectrum, Lifetime, $g^{(2)}$, Stability, and Saturation) associated with that coordinate.
- **Live Fitting:** The app utilizes the `src/fitting.py` module to perform on-the-fly biexponential decay fits and $g^{(2)}(0)$ dip calculations, providing immediate feedback on emitter purity and dynamics.

7.4 Automated Report Generation (`generate_report.py`)

For autoscan characterization runs, the `generate_report.py` script provides batch-processing. When ran, the script scans the uploaded data files, groups all files by their coordinate metadata, and generates a multi-page PDF summary (`Emitter_Report.pdf`). Each page is dedicated to a unique emitter, providing a standardized summary of its photophysical properties. This ensures that every point from an Autoscan is documented with publication-ready plots and tabulated metadata with minimal effort. An overnight autoscan can easily result in 1000 characterized emitters, depending on integration time settings, and this report is useful for visualizing that data in a convenient way.

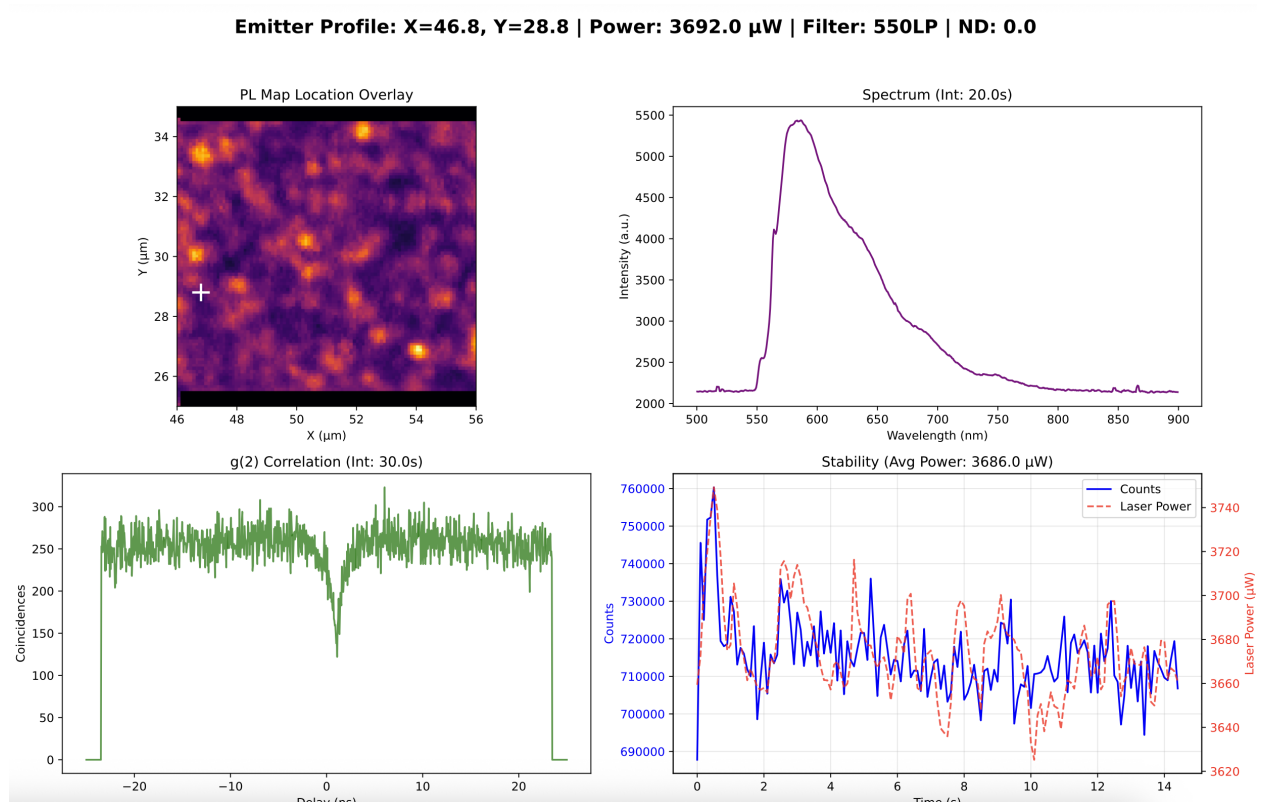


Figure 26: A sample page from a generated report. Data shown is an emitter discovered in our group that is hosted within silicon nitride.

7.5 Software Structure

The analysis repository is organized into modular components:

- `src/parsers.py`: Logic for regex-based metadata extraction.
- `src/measurements.py`: Classes for loading and normalizing each data type.
- `src/fitting.py`: Physics-based models for lifetime deconvolution, $g^{(2)}$ antibunching fits, and power saturation curves.

Users can use these core files that to build their own apps or analyses.

8 Troubleshooting

Error / Symptom	Solution
Crash upon start up	Check that the stage is on.
No counts on SPCM	Check that the power supply for it is on. Must be turned on manually.
Significant fast oscillations in the count traces	Change the power meter integration time to a higher value, then retry it to desired value.
PL Map is blank or pure noise	Check the arduino connections (Section 2.3) and that the MATLAB server is open in the background.

Table 14: Common Troubleshooting Steps

Appendix: Acronym Definitions

This table provides a comprehensive list of the acronyms and technical abbreviations used throughout this manual.

Table 15: Glossary of Acronyms and Technical Terms

Acronym	Full Term	Context / Description
Timing & Correlation		
ADC	Analog-to-Digital Converter	Converts the amplified TAC voltage into a digital time tag.
AMP	Biased Amplifier	Selects and amplifies a specific time window within the TAC range.
CFD	Constant Fraction Discriminator	High-precision trigger used to eliminate amplitude-dependent timing jitter.
HBT	Hanbury Brown and Twiss	Experimental configuration used for $g^{(2)}$ photon correlation.
IRF	Instrument Response Function	The total timing uncertainty/broadening of the measurement system.
TAC	Time-to-Amplitude Converter	Core TCSPC component that converts start-stop intervals into voltage.
TCSPC	Time-Correlated Single-Photon Counting	The primary statistical technique for lifetime and correlation data.
TTS	Transit Time Spread	The internal timing jitter of a photon detector.
Optics & Hardware		
AOM	Acousto-Optic Modulator	High-speed crystal used for laser pulsing and power control.
CW	Continuous-Wave	Refers to a laser operating in constant (non-pulsed) mode.
NA	Numerical Aperture	Measure of the light-gathering capability of the objective lens.
NV	Nitrogen-Vacancy	The specific type of quantum emitter (point-defect) in diamond.
PBS	Polarizing Beam Splitter	Optical component used to route light based on polarization state.
PL	Photoluminescence	The light emission collected from the sample after excitation.
QWP	Quarter-Wave Plate	Retardation plate used to convert linear polarization to circular.
RSP	Right Side Port	The specific physical output port on the Nikon Ti-U microscope.
Detectors & Electronics		
DAQ	Data Acquisition	Hardware interface (NI USB-6341) for counting and control.
NIM	Nuclear Instrumentation Module	Fast, current-based signaling standard (~ -800 mV) for timing.
PDM	Photon Detection Module	High-speed SPAD optimized for sub-ps timing resolution.
PFI	Programmable Function Interface	Configurable digital terminals on the NI DAQ device.
SPAD	Single-Photon Avalanche Diode	The class of detector used for counting individual photons.
SPCM	Single-Photon Counting Module	SPAD optimized for high efficiency and general PL scanning.
TTL	Transistor-Transistor Logic	Voltage-based digital signaling standard (~ 5 V) for counting.

References

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